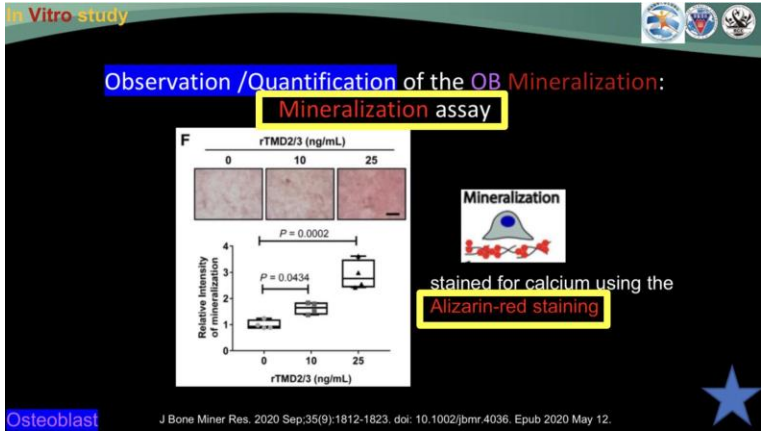


BM111 BLOCK-8 期中考詳解

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骨質疏鬆與骨壞死	林松彥
骨科學研究的細胞生理與動物模式(實驗)	鄭琮霖
手部、腕部骨科功能解學	邱兆澤
脊椎之骨科功能解剖學	高志銘
肌肉骨骼免疫學	李英瑞
骨折的生物力學及癒合	李天慶
基礎影像學原理	蔡欣學
肌肉骨骼胚胎學	何宛怡
肩肘關節之骨科功能解剖學/骨科鑑別診斷	盧政昌
足踝關節之骨科功能解剖學	陳姝蓉
骨骼組織學	施耀翔
組織學-肌腱、韌帶、肌肉	陳永佳
肌肉骨骼病理學(1)	陳建亨
肌肉疾病	梁文貞
非類固醇抗炎藥物之藥理學	吳炳男
肌肉骨骼影像解剖學(1)-上肢	林育全
免疫風溼學實驗室檢查	曾家駿
骨與軟骨生理學	何美玲
生理學實驗-骨質疏鬆動物模式	何美玲
組織學與實驗-骨骼與軟骨	張瀛双
肌肉鬆弛劑之藥理學	程廣義
基礎生物力學	周伯禧
運動創傷及運動醫學	周伯禧
組織學與實驗-肌肉、肌腱、韌帶	劉俊馳
骨科學復健(I)	李佳玲
骨科學復健(II)	曾昱璇

1	Which of the following methods is more suitable for observing the mineralization ability of osteoblasts? (A) Alizarin-Red staining (B) Jennifer-Yellowing staining (C) Turkiye-blue staining (D) Hermès-Green staining
答案	(A) Alizarin-Red staining
詳解	 In Vitro study Observation / Quantification of the OB Mineralization: Mineralization assay Mineralization stained for calcium using the Alizarin-red staining Osteoblast J Bone Miner Res. 2020 Sep;35(9):1812-1823. doi: 10.1002/jbmr.4036. Epub 2020 May 12.
出處	鄭琮霖 骨科學研究的細胞生理與動物模式(實驗)

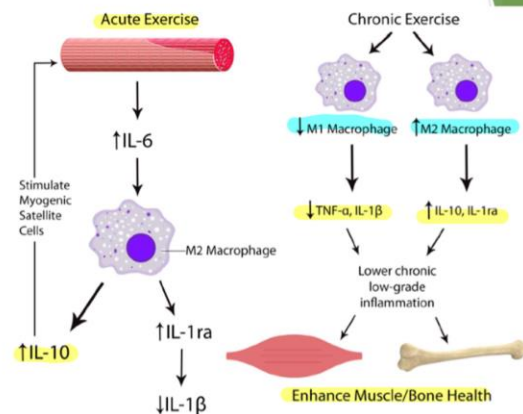
2	What is incorrect regarding the functions of Macrophage in wound healing and fracture healing? (A) It can promote and resolve inflammation of the healing process. (B) It can remove death cells. (C) It can provide signal molecules for healing process. (D) It can provide factors to against angiogenesis.															
答案	(D) It can provide factors to against angiogenesis.															
詳解	Table 1. The role of macrophages in the different wound-healing phases <table><tr><th>Phase</th><th>Cells involved</th><th>Function and activity</th></tr><tr><td>Haemostasis</td><td>Platelets</td><td>Clotting</td></tr><tr><td>Inflammation</td><td>Macrophages Neutrophils</td><td>Tissue macrophages • alarms and attracts neutrophils to wound bed • phagocytise debris • produce cytokines and growth factors </td></tr><tr><td>Proliferation</td><td>Macrophages Fibroblasts Myofibroblasts Keratinocytes</td><td>Macrophages • produce signal molecules for angiogenesis • activate fibroblasts to collagen production • attract and activate myofibroblasts for wound contraction • produce growth factors for cell proliferation </td></tr><tr><td>Remodelling</td><td>Fibroblasts Macrophages Fibrocytes</td><td>Fibrocytes are "clones" of macrophages and fibroblasts, aiding collagen-tissue modulation</td></tr></table> (D) 由表格中得知，在Proliferation Phase，Macrophage會產生信號以進行angiogenesis.	Phase	Cells involved	Function and activity	Haemostasis	Platelets	Clotting	Inflammation	Macrophages Neutrophils	Tissue macrophages • alarms and attracts neutrophils to wound bed • phagocytise debris • produce cytokines and growth factors	Proliferation	Macrophages Fibroblasts Myofibroblasts Keratinocytes	Macrophages • produce signal molecules for angiogenesis • activate fibroblasts to collagen production • attract and activate myofibroblasts for wound contraction • produce growth factors for cell proliferation	Remodelling	Fibroblasts Macrophages Fibrocytes	Fibrocytes are "clones" of macrophages and fibroblasts, aiding collagen-tissue modulation
Phase	Cells involved	Function and activity														
Haemostasis	Platelets	Clotting														
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Proliferation	Macrophages Fibroblasts Myofibroblasts Keratinocytes	Macrophages • produce signal molecules for angiogenesis • activate fibroblasts to collagen production • attract and activate myofibroblasts for wound contraction • produce growth factors for cell proliferation														
Remodelling	Fibroblasts Macrophages Fibrocytes	Fibrocytes are "clones" of macrophages and fibroblasts, aiding collagen-tissue modulation														

出處	李英瑞 肌肉骨骼免疫學
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3	Immunity dysregulation is an important issue for sarcopenia. what is correct in the following? (A) The immunity in the old persons is high comparing young person, which will elevate chronic infections to induce a chronic inflammation to enhance the risk of sarcopenia (B) The overexpression IL-6 can be found in the patients with sarcopenia to reduce immunity. (C) Exercise can provide benefit for patients with sarcopenia to reduce chronic inflammation and slow the progression. (D) Elevating M1 macrophage can reduce chronic inflammation and help muscle regeneration.
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答案	(C) Exercise can provide benefit for patients with sarcopenia to reduce chronic inflammation and slow the progression.
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詳解	◎ Sarcopenia 肌少症 一、肌少症的起因與致病過程 1. 造成 sarcopenia 肌少症的原因有很多，雖然 inflammation 不是最重要的但是有越來越多研究顯示 inflammation 是不可避免的。 2. 隨著年紀增長免疫會越發低下造成 chronic infection 的機會增加導致身體會處於 low level 的慢性發炎狀態，隨著時間累積這種發炎狀態會使肌少症的產生。 3. (M107) 起因：營養不足、體內發炎、胰島素抗阻、賀爾蒙失調、運動量不足、維生素 D 不足、粒線體障礙。 4. (M107) 身體持續性的慢性發炎會導致蛋白質的分解，造成肌肉細胞的破壞。以上原因導致肌肉合成減少、神經退化，便會產生肌肉不足或肌肉無力的現象。 A選項應該是因為老人免疫會低下，造成身體處於低度發炎反應隨時間累積而造成肌少症。
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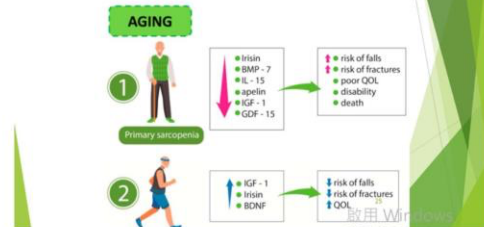


1. 在慢性發炎中是由 M1 macrophage 來主導，如果可以使 M1 macrophage 減少 M2 macrophage 增加就可以避免慢性發炎的產生。其中運動是一個很好的方法。
M1 ↓, M2 ↑

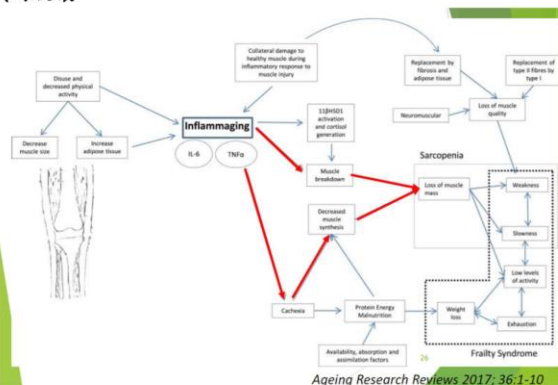
D 選項應該改為 M2 增加

二、老化發炎(inflammaging)和肌少症的關係

Inflammaging, which is characterized by **increased** levels of proinflammatory cytokines such as **IL-1-β, IL-6, and TNF-α** as well as **CRP**, and a **reduced** serum level of anti-inflammatory cytokines, plays a central role within the creation and maintenance of sarcopenia states. The link between sarcopenia and immunosenescence could be explained by the increased systemic inflammation, with aberrant neutrophil migration potentially contributing to inflammaging and tissue damage.



1. 老化發炎(inflammaging)會導致 proinflammatory cytokines 的量增加，如 IL-1-β、IL-6、TNF-α、CRP，以及 anti-inflammatory cytokines 減少。
2. Proinflammatory cytokines 的增加會導致嗜中性球的大量活化與聚集，造成組織的破壞。



3. IL-6、TNF-α 在 sarcopenia 扮演很重要的角色，因此在治療 sarcopenia 有時候會用 anti-TNF-α 來治療但效果 case by case。

C 選項應該是 induce inflammatory

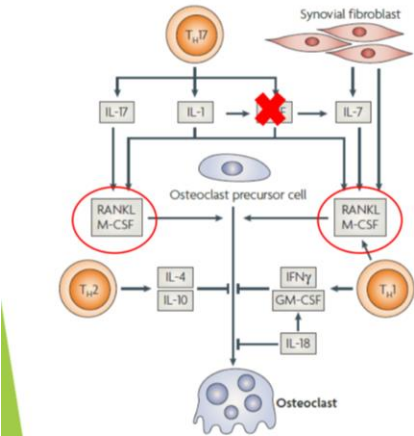
出處

李英瑞 肌肉骨骼免疫學(共筆pg. 25, 26)

4	Immune-regulation is one of the factors of rheumatoid arthritis. What is incorrect in the following? (A) Autoimmunity isn't the issue for rheumatoid arthritis. (B) TH17 cells will release IL-1 and IL-17 and promote chondrocytes from matrix synthesis to be matrix degradation type, and elevating rheumatoid arthritis. (C) Tumor-necrosis factor (TNF) will promote osteoclast differentiation and contributing on rheumatoid arthritis. (D) Numbers of environmental and genetic factors are associated with immune-regulation and disease incidence.
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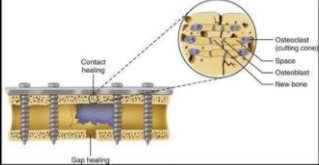
答案	(A) Autoimmunity isn't the issue for rheumatoid arthritis.
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詳解	(A) RA為自體免疫疾病 (D) 促成RA發生的危險因子包含環境及基因因素 Rheumatoid Arthritis ▶ a chronic inflammatory disease ▶ targets the synovial membrane, cartilage and bone ▶ affects 1% of the population ▶ women between 30-50 years old ▶ numbers of environmental and genetic risk factors associated with disease incidence (smoking, obesity, vitamin D deficiency, changes in the microbiota, HLA type, etc.) ▶ Numerous cytokines are expressed and are functionally activation. ▶ induction of autoimmunity, chronic inflammation and thereby joint damage (B) Th17所分泌的IL-1及IL-17會誘導軟骨細胞由合成基質轉為分解基質的型態 Pathways regulating chondrocyte activation and cartilage degradation in rheumatoid arthritis. Cartilage degradation is a multistep process based on the release of matrix-degrading enzymes such as aggrecanases (ADAMTS) and matrix metalloproteinases (MMPs). Cytokines such as interleukin-1 (IL-1) and IL-17 induce a switch in the synthesis pattern of chondrocytes from matrix molecules to matrix-degrading enzymes. In addition, synovial fibroblasts start producing matrix-degrading enzymes and invade cartilage when activated by cytokines such as tumour necrosis factor (TNF) and IL-1. Chondrocyte death is another feature of cartilage damage, it leads to the formation of empty lacunae and deprives cartilage from the ability to replenish matrix. (C) IL-1、TNF、IL-17、IL-7在osteoclast的分化上扮演重要角色
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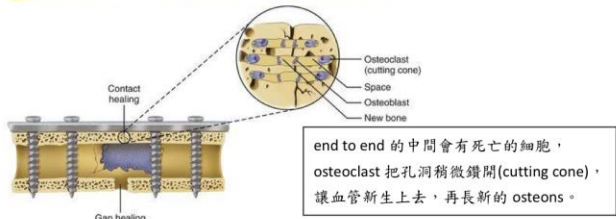
	 The key factors that regulate osteoclast differentiation in rheumatoid arthritis. The differentiation of bone-resorbing osteoclasts from haematopoietic osteoclast precursors is a cytokine-driven process. The essential cytokine mediators are RANKL (receptor activator of nuclear factor-κB (RANK) ligand) and M-CSF (macrophage colony-stimulating factor), which are expressed by synovial fibroblasts and T helper 1 (TH1) cells. Osteoclast differentiation is achieved by the actions of tumour-necrosis factor (TNF) and interleukin-1 (IL-1), as well as of IL-17, produced by TH17 cells, and IL-7, produced by synovial fibroblasts. By contrast, IL-4 and IL-10, which are produced by TH2 cells, and granulocyte/M-CSF (GM-CSF) and interferon-γ (IFNγ), which are produced by TH1 cells, inhibit osteoclast differentiation.
出處	李英瑞 肌肉骨骼免疫學

5	What factor would affect the blood supply of fractures? (A) The mechanism of injury (B) Early fixation (C) The choice of surgical approaches (D) All of above
答案	(D) All of above
詳解	Fracture and blood supply Factors influence blood supply • Mechanism of injury • Initial patient management – early fixation • Patient resuscitation – hypovolemia and hypoxia • Surgical approach • Implant • Consequences of trauma 
出處	李天慶 骨折的生物力學與癒合(ppt pg. 7)

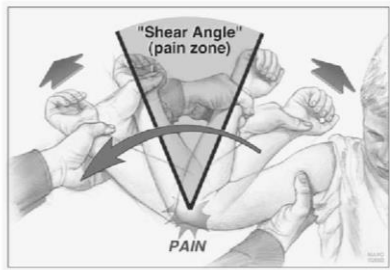
6	Which statement about the types of fracture healing is CORRECT? (A) The primary bone healing depends on relative stability. (B) The secondary bone healing needs absolute stability to induce callus formation. (C) The most common type of fracture healing is secondary bone healing. (D) The secondary bone healing is a process of osteonal bone remodeling.
答案	(C) The most common type of fracture healing is secondary bone healing.

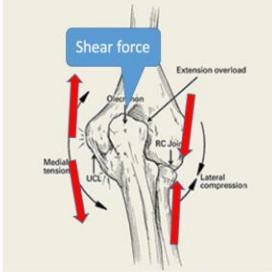
詳解	(A) The primary bone healing depends on absolute stability. (B) 需要absolute stability的是primary bone healing。 secondary healing只要relative stability就好 (D) osteonal bone remodeling是primary bone healing走的路徑。 Block 8: Musculoskeletal System 高雄醫學大學 Kaohsiung Medical University Types of fracture healing • Primary (direct healing): by internal remodeling • absolute stability • osteonal bone remodeling: Cutting cones -> angiogenesis -> new osteons • Secondary (indirect healing): by callus formation • relative stability • both intramembraneous and endochondral bone formation  11
出處	李天慶 骨折的生物力學與癒合(ppt pg. 11)

7	Which kind of implants would “least” cause the primary bone healing? (A) Intramedullary nails (B) Tension band wires (C) Compression plates (D) Buttress plates
答案	(A) Intramedullary nails
詳解	三、Surgical fixation with relative stability 相對穩定性時的手術治療 1. Displacement increases with the applied load and decreases with the rigidity of the fixation device 利用固定裝置使移位減少，僅在合理的範圍內做 micromotion 2. There is no exact definition of the required or tolerated flexibility 選擇何種固定裝置及如何鎖都會影響到穩定性 3. All fixation methods with the exception of compression techniques may be seen as flexible fixations 任何骨折的固定若沒有使骨塊 compression 的情況下都不能稱作絕對穩定，因為都可能有相對性位移。 4. <u>Implants: external fixators, intramedullary nails, or internal fixators</u> 常見固定方式為外固定、鋼釘、內固定

	1. Primary (direct healing): by internal remodeling 骨折癒合比較少走的路徑，癒合條件較嚴苛，過程不經由骨痂，需要 bone to bone contact，end to end 要做得很好。  end to end 的中間會有死亡的細胞，osteoclast 把孔洞稍微鑽開(cutting cone)，讓血管新生上去，再長新的 osteons。 (1) absolute stability 固定非常牢靠 (2) osteonal bone remodeling: primary bone healing 又叫做 osteonal healing A. Cutting cones -> angiogenesis -> new osteons 2. Secondary (indirect healing): by callus formation 絕大部分骨折癒合走的路徑，經由骨痂來癒合 (1) relative stability 比較不穩定 (2) both intramembraneous and endochondral bone formation 膜內骨化與軟骨內骨化並行 Primary bone healing 要求的是 absolute stability，secondary bone healing 則是 relative stability，由共筆可見能提供 relative stability 的 surgery fixation 是 intramedullary nail, external fixators, and internal fixators。
出處	李天慶 骨折的生物力學與癒合

8	Miss Lin is a volleyball player in KMU. She sustained right shoulder pain after shoulder overhead motion during volleyball training. Which statement is not right? (A) The image examination of X ray should include shoulder AP view, scapular Y view or axillary view (B) The long head of biceps is inserted and connected with superior labrum of shoulder, which might injury in overhead motion (C) Rotator cuff injury is not possible in this case (D) To survey the soft tissue injury in shoulder joint, MRI is preferred
答案	(C) Rotator cuff injury is not possible in this case
詳解	盧老師的考題愛考肩膀疾病的鑑別診斷，一定要背列入診斷考量的疾病種類，包括 Bone, Cartilage, Capsule and ligament, Bursa, Tendon, Nerve 這幾種。 A 選項為肩膀 X ray 常見從不同角度拍攝的種類可以參考影像學有相關內容，這幾個角度可以協助從不同方向看肩膀的病灶 BC 考的是肩膀的 tendon 常見的鑑別診斷，B 是正確且 biceps tendon 的兩個 proximal insertion 常考（短頭：coracoid process/長頭：supraglenoid tubercle） D 正確，MRI 對軟組織鑑別度較好，可以用在肩膀疾病鑑別
出處	盧政昌 肩肘關節之骨科功能解剖學/骨科鑑別診斷

9	Mr. Wang is a pitcher of Chinese Professional Baseball League, he complained of right elbow medial side pain after throwing, which statement is “not” right? (A) Ulnar nerve function should be checked. (B) “Tennis elbow” should be first considered. (C) Physical examination should include the elbow valgus stress test. (D) The most common pitcher’s ligament injury of elbow is ulnar collateral ligament injury.		
答案	(B) “Tennis elbow” should be first considered.		
詳解	(A) Ulnar nerve 在手臂內側，因此也要一併確認。 (B) Tennis elbow是lateral epicondylitis，而Mr. Wang是medial side elbow pain，因此為錯誤選項。 <i>Tendon</i> <table border="0"> <tr> <td style="text-align: center;"> Tennis elbow Lateral epicondylitis  LATERAL ELBOW Problem occurs due to repetitive stress which causes interstitial tearing, inflammation and swelling. Extensor muscle insertion </td> <td style="text-align: center;"> Golf elbow Medial epicondylitis  MEDIAL ELBOW Pronator and Flexor muscle insertion </td> </tr> </table> Tennis Elbow / Golfer's Elbow (C) Special test – Moving valgus stress test  "Shear Angle" (pain zone) PAIN 1. Externally rotates the glenohumeral joint and applies a valgus force on the elbow 2. Check stability at approximately 30° while on the joint, noting the position(s) in which pain is evoked. 3. Maintain a valgus force and move the elbow from extension into flexion Dx. Instability of the medial capsular ligament (D)	Tennis elbow Lateral epicondylitis  LATERAL ELBOW Problem occurs due to repetitive stress which causes interstitial tearing, inflammation and swelling. Extensor muscle insertion	Golf elbow Medial epicondylitis  MEDIAL ELBOW Pronator and Flexor muscle insertion
Tennis elbow Lateral epicondylitis  LATERAL ELBOW Problem occurs due to repetitive stress which causes interstitial tearing, inflammation and swelling. Extensor muscle insertion	Golf elbow Medial epicondylitis  MEDIAL ELBOW Pronator and Flexor muscle insertion		

	四、Valgus Extension Overload, Osteophyte Impingement, OCD 外翻伸展過度，骨刺夾擠（重點） 1. Late cocking-acceleration 投手在做上肩型投球時，手肘呈現一個很大的外翻伸展的動作 (1) 內側(ulnar side)：tension force (2) 外側(radial side)：compression force (3) 後方(olecranon fossa)：shear force（剪力）   2. 內側的 tension force 不斷地拉扯，一次一次累積損害，久了就會造成 ulnar collateral ligament 斷掉，投手球速下降與此有關。 3. 治療：Tommy John Procedure (ulnar collateral ligament reconstruction)，重建損毀的韌帶
出處	盧政昌 肩肘關節之骨科功能解剖學/骨科鑑別診斷

10	Mr. Chen complained of right shoulder pain after work. He visited an Orthopedic clinic, which items should consider to make the differential diagnosis of shoulder pain? a. X-ray b. Physical examination c. Income d. Patient's age e. Occupation f. Shoulder anatomy g. Injury history (A) a,b,c,d,e,f (B) a,c,d (C) a,b,d,e,f,g (D) a,b,c,e,f
答案	(C) a,b,d,e,f,g

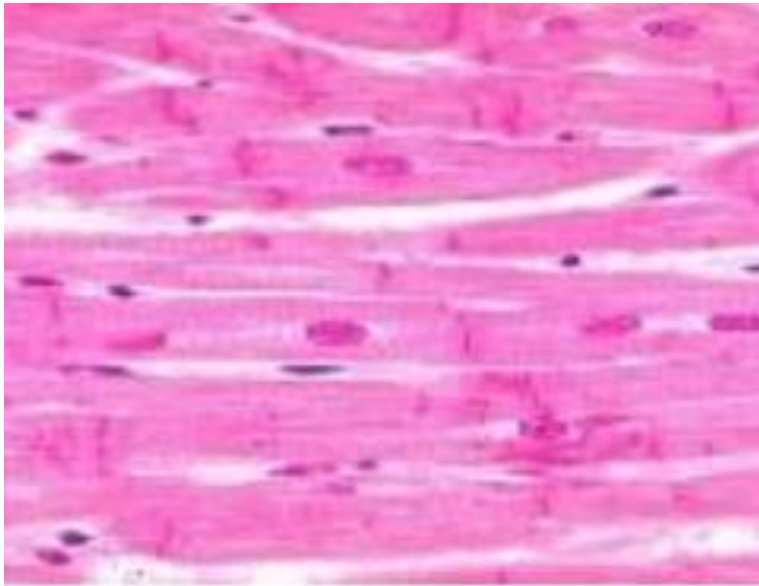
詳解	◎ Differential Diagnosis in Orthopaedic 一、介紹（老師跳過） 1. Anatomy – Associated Disease 2. Disease / Symptoms sign 3. Disease Factors: Age, Occupation, Injury Mechanism, Others factors 好發的年紀、職業、受傷機轉等等 4. Physical examination 理學檢查 5. Image study 影像學檢查 6. First impression 7. 做以上診斷的主要目的是為了治療病人，而治療又分為保守治療，例如吃藥、打針、復健等非侵入性的治療，如果有明顯的損傷，則需要進行手術
出處	盧政昌 肩肘關節之骨科功能解剖學/骨科鑑別診斷

11	About ankle and foot, which one is correct? (A) The cause of Muller Weiss disease is osteonecrosis of navicular bone. The one character of this disease is comma-shape deformity of navicular bone (B) The talus is the keystone of transverse arch of foot (C) The peroneal longus tendon attaches to 5th metatarsal base and everts the foot (D) In the syndesmosis, the tibia is concave and fibular is convex here. The anterior talofibular ligament and posterior talofibular ligament are part of the syndesmosis ligaments
答案	(A) The cause of Muller Weiss disease is osteonecrosis of navicular bone. The one character of this disease is comma-shape deformity of navicular bone
詳解	(B) 應為navicular (C) 附著在first metatarsal bone (D) Sydesmosis ligaments 有anterior inferior tibiofibular ligament, posterior inferior tibiofibular ligament, interosseous ligament
出處	陳妹蓉 足踝關節之骨科功能解剖學

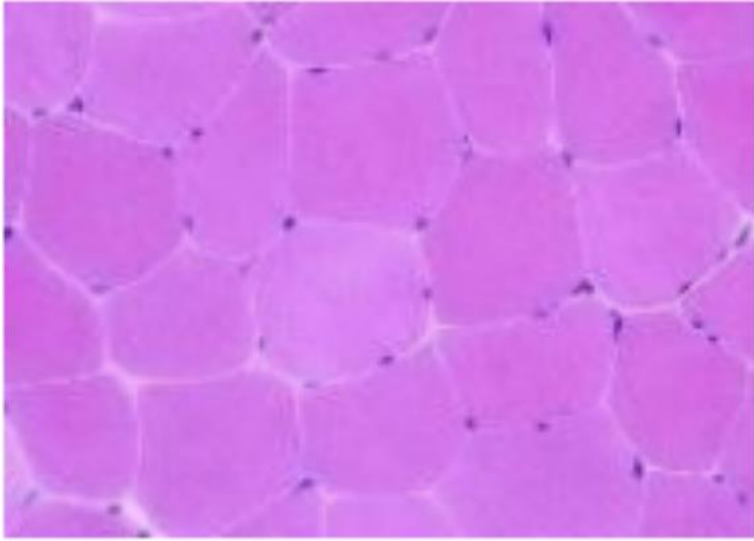
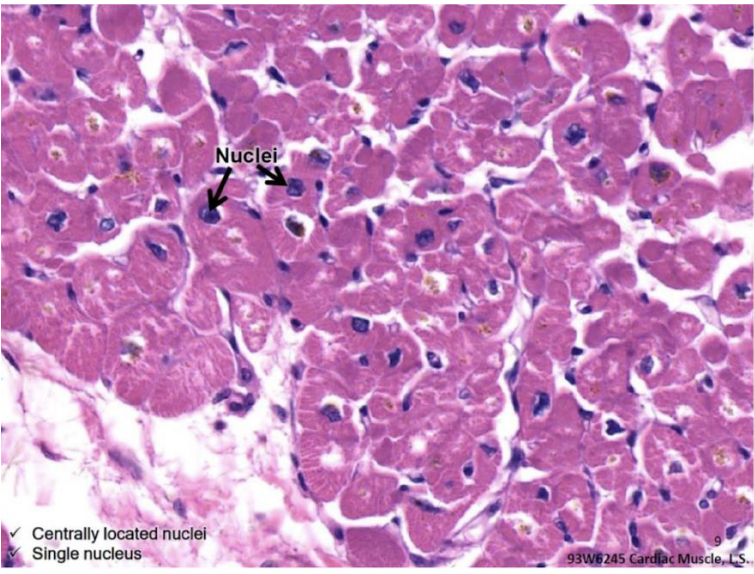
12	About the four compartments of leg, which one is wrong? (A) There are major neurovascular bundles in each compartment, except superficial posterior compartment. (B) The major movement of muscles in anterior compartment is ankle dorsiflexion. (C) There are deep peroneal nerve, peroneal longus and brevis in the lateral compartment. The major movement of these muscles is foot eversion. (D) The plantaris is in superficial posterior compartment and it is a weak ankle plantarflexor.
答案	(C) There are deep peroneal nerve, peroneal longus and brevis in the lateral compartment. The major movement of these muscles is foot eversion.

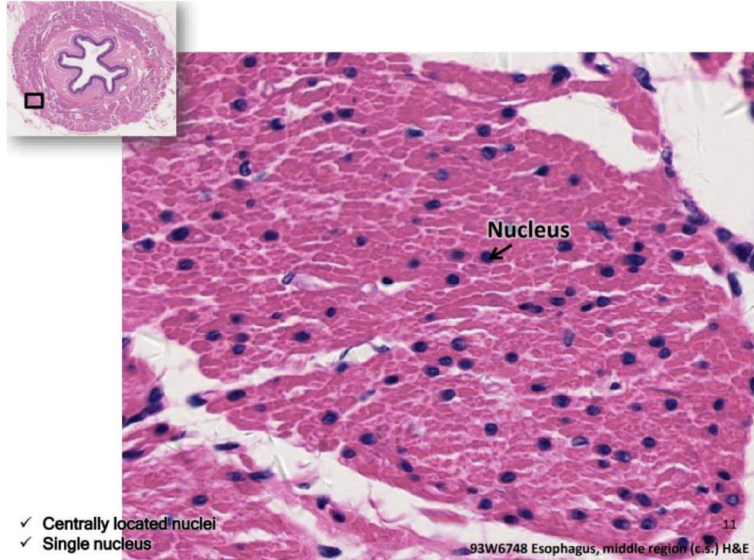
詳解	Lateral compartment所含的神經應為 superficial peroneal n.
出處	陳姝蓉 足踝關節之骨科功能解剖學

13	Which one is wrong? (A) Near 50% of blood supply to talus is from posterior tibial artery, which is in the deep posterior compartment. (B) The common complication of displaced talar neck fracture is osteonecrosis of talus, (C) The anterior talofibular ligament is most easily injured ligament when patients suffer from ankle sprain, and anterior drawer test is often positive while performing physical examination. Repeated ankle sprains may lead to lateral instability of ankle joint. (D) The negative Thompson test means Achilles tendon rupture. Under this condition, patient totally can't walk therefore.
答案	(D) The negative Thompson test means Achilles tendon rupture. Under this condition, patient totally can't walk therefore.
詳解	Thompson test (+) 才是 Achilles tendon rupture。
出處	陳姝蓉 足踝關節之骨科功能解剖學

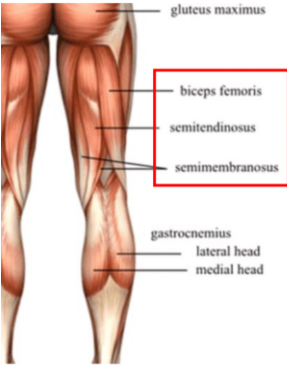
14	Observe the histological features. What is the tissue in this figure?  (A) Cardiac muscle (B) Tendon (C) Smooth muscle (D) Skeletal muscle
答案	(A) Cardiac muscle

詳解	看到 intercalated disc -> 心肌很重，所以是心肌
出處	劉俊馳 組織學與實驗-肌肉、肌腱、韌帶

15	What is the correct combination of tissue species and section orientation in the figure?  (A) cross sectioned cardiac muscle (B) longitudinal sectioned smooth muscle (C) cross sectioned smooth muscle (D) cross sectioned skeletal muscle
答案	(D) cross sectioned skeletal muscle
詳解	從切片可以知道是cross/transverse section，先排除B 心肌細胞的細胞核旁邊會有一些被洗掉的空腔，如下圖。  平滑肌細胞細胞核位於中央，也旁邊會不會有空腔，如下圖。

	 因此，從細胞型態可以知道是骨骼肌。
出處	劉俊馳 組織學與實驗-肌肉、肌腱、韌帶

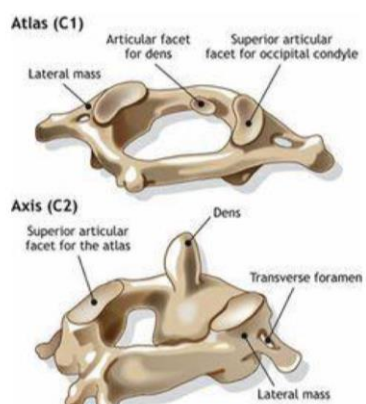
16	Which option is not suggested for management of soft tissue injury in acute phase? (A) Elevation (B) Protection (C) Vascularization (D) Compression																								
答案	(C) Vascularization																								
詳解	★ <i>Management-acute soft tissue injury</i> <table border="1" style="margin: 10px auto;"> <thead> <tr> <th colspan="2">Acute phase</th> </tr> </thead> <tbody> <tr><td>P</td><td>Protection</td></tr> <tr><td>E</td><td>Elevation</td></tr> <tr><td>A</td><td>Avoid</td></tr> <tr><td>C</td><td>Compression</td></tr> <tr><td>E</td><td>Education</td></tr> </tbody> </table> & <table border="1" style="margin: 10px auto;"> <thead> <tr> <th colspan="2">Subacute phase</th> </tr> </thead> <tbody> <tr><td>L</td><td>Load</td></tr> <tr><td>O</td><td>Optimism</td></tr> <tr><td>V</td><td>Vascularisation</td></tr> <tr><td>E</td><td>Exercise</td></tr> <tr><td></td><td></td></tr> </tbody> </table> 5	Acute phase		P	Protection	E	Elevation	A	Avoid	C	Compression	E	Education	Subacute phase		L	Load	O	Optimism	V	Vascularisation	E	Exercise		
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出處	曾昱璇 骨科學復健(2)(ppt pg. 5)																								

17	Which description as below about hamstring injury is incorrect? (A) We can use sonography for grading the severity of hamstring injury. (B) Acute overload happens when the muscle is near full flexion or full extension. (C) Hamstring is usually injured when forceful concentric contractions. (D) The strength ratio should improve to an isokinetic strength ratio of 0.6 hamstrings to quadriceps before return to most sports.
答案	(C) Hamstring is usually injured when forceful concentric contractions.
詳解	Hamstring is usually injured when forceful eccentric contractions. Hamstring injury  Injury mechanism: - Usually injured when forceful eccentric contractions - Acute overload when the muscle is near full flexion or full extension <pre> graph TD A[Injury mechanism] --> B[Ice, gentle ROM exercise] B --> C[Grading the severity of injury] C --> D[Isokinetic strength ratio hamstrings: quadriceps ≥ 0.6] D --> E[Return to most sports] </pre> 25 ★
出處	曾昱璇 骨科學復健(2)(ppt pg. 25)

18	Which description as below about finger flexor tendon rupture is incorrect? (A) Zone III is called as no man's land. (B) Zone IV is within the carpal tunnel. (C) In acute phase, dorsal blocking splint is usually prescribed for positioning. (D) Dorsal blocking splint limits the amount of active and passive finger extension to prevent possible tendon rupture.
答案	(A) Zone III is called as no man's land.

詳解	Finger flexor tendon rupture Zone classification of injuries to the flexor tendons Zone I contains flexor digitorum profundus only, and is from its insertion to the insertion of flexor digitorum superficialis. Zone II (once known as 'no-man's land') is from the insertion of flexor digitorum superficialis to the proximal edge of the A1 pulley. Zone III is in the palm from the proximal end of A1 to the distal edge of carpal tunnel, and contains the origin of the lumbrical muscles. Zone IV is within the carpal tunnel. Zone V is proximal to the carpal tunnel in the forearm. Positioning Range of motion exercise Massage for reducing edema Tendon gliding exercise Strengthening exercise Endurance exercise Return to work Fig. 15.48 Open flexor tendon repair orthosis used for postoperative care. (Courtesy, Jeanne Davis, DPT, CMT) 30
出處	曾昱璇 骨科學復健(2)(ppt pg. 30)

19	Which of the following statements about the intervertebral disc is not true? (A) Nucleus pulposus has large proportion of hydrophilic proteoglycans. (B) Nutrients of the intervertebral disc are mainly obtained by passive diffusion through the end plate. (C) The intervertebral discs are distributed between each level of vertebra and account for approximately 20% to 33% of the spine height. (D) The composition of annulus fibrosus includes type I collagen and fibrocartilage, and its main function is to limit the motion of the vertebral body.
答案	(C) The intervertebral discs are distributed between each level of vertebra and account for approximately 20% to 33% of the spine height.
詳解	(C) 並不是每個椎體間都有椎間盤，例如C1和C2間，S2之後沒有 Intervertebral Disk Function ◦ Stability ◦ Resisting loads in all directions ◦ Restricting intervertebral motion Twenty-three intervertebral discs ◦ starting between C2 and C3 ◦ extending distally to L5-S1 Account for roughly 20% to 33% of the vertebral column height 14
出處	高志銘 脊椎之骨科功能解剖學(ppt pg. 14)

20	Which of the following statements about the structure of spine is false? (A) The vertebral body is the weight-bearing structure that exists in front of each level of vertebra. (B) The articular surface of the facet joint is covered with hyaline cartilage. (C) Dens is a special structure of axis, which is part of the Atlantoaxial joint. This joint is responsible for cervical rotation. (D) The ribs and thoracic vertebrae are connected to each other by costal facets, which restricts the movement of the spine. Therefore, the thoracic spine is the least mobile part of the spinal column.
答案	(A) The vertebral body is the weight-bearing structure that exists in front of each level of vertebra.
詳解	(A) C1並沒有 true ventral body Cervical Vertebrae The first cervical vertebra (Atlas) - unique in its articulation with the occipital condyle - it lacks a true ventral body - supports the cranium by the superior facet surfaces of the lateral masses  21
出處	高志銘 脊椎之骨科功能解剖學(ppt pg. 21)

21	Which part of spine does not belong to the posterior ligamentous complex? (A) ligamentum flavum (B) interspinous ligament (C) posterior longitudinal ligament (D) facet joint capsule
答案	(C) posterior longitudinal ligament
詳解	posterior longitudinal ligament 屬於 middle column

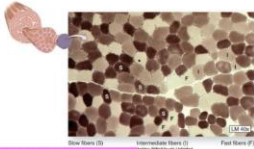
	Ligament & Column 20
出處	高志銘 脊椎之骨科功能解剖學(ppt pg. 20)

22	The structure terminal cisternae are formed by? (A) thick filaments (B) thin filament (C) enlarged sarcoplasmic reticulum (D) invagination of sarcoplasm
答案	(C) enlarged sarcoplasmic reticulum
詳解	Comparison of sER system in cardiac and skeletal muscle • Poorly organized sER system - Diad: small terminal cisternae + T tubule The T tubules are larger and more numerous in cardiac ventricular muscle than in skeletal muscle. Figure 10.7. Diagram of the organization of striated muscle fiber. ppt 中可以看到terminal cisternae為sarcoplasmic reticulum的一部分。 33
出處	陳永佳 組織學-肌腱、韌帶、肌肉(ppt pg. 33)

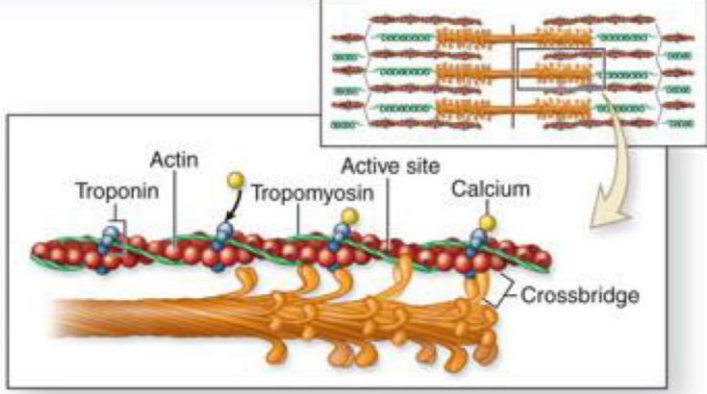
23	Which of the following description is false? (A) When injury, the skeletal muscle fibers and cardiomyocytes can undergo regeneration (B) All of the three types of muscle present gap junctions (C) Smooth muscle fibers do not present sarcomere (D) In normal conditions, only the smooth muscle fibers could undergo regeneration freely.																																								
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申覆	原因： 1. 根據老師的ppt pg. 45的表格，cardiomyocyte在regeneration一欄為NONE(Normal condition)。 2. 題目雖有說明為When injury，但根據2021年的期刊：Gong, R., Jiang, Z., Zagidullin, N. et al. Regulation of cardiomyocyte fate plasticity: a key strategy for cardiac regeneration. Sig Transduct Target Ther 6, 31 (2021). 目前成人心肌修復與再生的比例極低，且有諸多變數，足見尚未有定論，因此判斷(A)選項亦為錯誤敘述，在此附上引用資料截圖，望老師斟酌給分，謝謝老師！ 申覆結果：維持原答案 老師回覆： 因A選項雖如同學所說目前未有定論，但仍有不少證據支持cardiac regeneration的發現。Bergmann, O., Bhardwaj, R.D., Bernard, S., Zdunek, S., Barnabé-Heider, F., Walsh, S. et al. (2009) Evidence for cardiomyocyte renewal in humans. Science 324, 98–102 10.1126/science.1164680 [PMC free article] [PubMed] [CrossRef] [Google Scholar] Senyo, S.E., Steinhauser, M.L., Pizzimenti, C.L., Yang, V.K., Cai, L., Wang, M. et al. (2013) Mammalian heart renewal by pre-existing card																																								

	omyocytes. Nature 493, 433 – 436 10.1038/nature11682 [PMC free article] [PubMed] [CrossRef] [Google Scholar] Mollova, M., Bersell, K., Walsh, S., Savla, J., Das, L.T., Park, S.-Y. et al. (2013) Card iomyocyte proliferation contributes to heart growth in young human s. Proc. Natl Acad. Sci. U.S.A. 110, 1446 – 1451 10.1073/pnas.12146 08110 [PMC free article] [PubMed] [CrossRef] [Google Scholar] http://www.ncbi.nlm.nih.gov/pmc/articles/PMC9042388/#BST-50-269C3
出處	陳永佳 組織學-肌腱、韌帶、肌肉(ppt pg. 42)

24	Which of the following statements concerning smooth muscle is true? (A) Cells are fusiform shaped and communicate with each other by an adherent junction. (B) The cytoplasm of smooth muscle is eosinophilic and its nuclei are localized in the central of the cell. (C) The fibers have dense bodies but do not present well-developed smooth ER. (D) The connective tissue of smooth fibers also contains endomysium and perimysium.																																								
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出處	陳永佳 組織學-肌腱、韌帶、肌肉(ppt pg. 44)																																								

25	Which of the following skeletal muscle fibers tend to fatigue? (A) slow oxidative (B) intermediate (C) fast aerobic fibers (D) fast anaerobic fibers																																																
答案	(D) fast anaerobic fibers																																																
詳解	Types of Skeletal Muscle Fibers Skeletal muscle fibers - Are divided into three types - How they manufacture energy (ATP) - How quickly they contract Oxidative fibers—produce ATP aerobically Glycolytic fibers—produce ATP anaerobically by glycolysis  Copyright © The McGraw-Hill Companies, Inc. Permission is granted by reproduction of this image. Slow fibers (I) Intermediate fibers (IIa) Fast fibers (IIb) Training effects Intensive resistance training can convert fast glycolytic fibers into fast oxidative fibers. Table 10.4 Structural and Functional Characteristics of Different Types of Skeletal Muscle Fibers <table><thead><tr><th>Fiber Characteristic</th><th>Slow Fibers (Type I, Slow Oxidative)</th><th>Intermediate Fibers (Type IIa, Fast Aerobic)</th><th>Fast Fibers (Type IIb, Fast Anaerobic)</th></tr></thead><tbody><tr><td>ATP Use</td><td>Slowly</td><td>Quickly</td><td>Quickly</td></tr><tr><td>Capacity to Make ATP</td><td>Aerobic</td><td>Aerobic</td><td>Anaerobic</td></tr><tr><td>Concentration of Capillaries</td><td>Extensive</td><td>Extensive</td><td>Sparse</td></tr><tr><td>Color of Fibers</td><td>Red</td><td>Red</td><td>White (pale)</td></tr><tr><td>Contraction Velocity</td><td>Slow</td><td>Fast</td><td>Fast</td></tr><tr><td>Resistance to Fatigue</td><td>Highest</td><td>High</td><td>Low</td></tr><tr><td>Fiber Distribution</td><td>Found in greatest abundance in muscles of the trunk, especially postural muscles</td><td>Found in greatest abundance in muscles of the lower limbs</td><td>Found in greatest abundance in muscles of the upper limbs</td></tr><tr><td>Fiber Diameter</td><td>Smallest</td><td>Intermediate</td><td>Largest</td></tr><tr><td>Number of Mitochondria</td><td>Many</td><td>Many</td><td>Few</td></tr><tr><td>Amount of Myoglobin</td><td>Large</td><td>Large</td><td>Small</td></tr><tr><td>Primary Fiber Function</td><td>Endurance (e.g., marathon running), maintaining posture</td><td>Medium duration, moderate movement (e.g., walking, biking)</td><td>Short duration, intense movement (e.g., sprinting, lifting weights)</td></tr></tbody></table> rhabdomyolysis Damage to skeletal muscles caused by crushing or by extreme exercise may lead to a condition called rhabdomyolysis. 肌紅蛋白可攜氧 由圖可知 fast anaerobic fiber 對 fatigue resistance 最低	Fiber Characteristic	Slow Fibers (Type I, Slow Oxidative)	Intermediate Fibers (Type IIa, Fast Aerobic)	Fast Fibers (Type IIb, Fast Anaerobic)	ATP Use	Slowly	Quickly	Quickly	Capacity to Make ATP	Aerobic	Aerobic	Anaerobic	Concentration of Capillaries	Extensive	Extensive	Sparse	Color of Fibers	Red	Red	White (pale)	Contraction Velocity	Slow	Fast	Fast	Resistance to Fatigue	Highest	High	Low	Fiber Distribution	Found in greatest abundance in muscles of the trunk, especially postural muscles	Found in greatest abundance in muscles of the lower limbs	Found in greatest abundance in muscles of the upper limbs	Fiber Diameter	Smallest	Intermediate	Largest	Number of Mitochondria	Many	Many	Few	Amount of Myoglobin	Large	Large	Small	Primary Fiber Function	Endurance (e.g., marathon running), maintaining posture	Medium duration, moderate movement (e.g., walking, biking)	Short duration, intense movement (e.g., sprinting, lifting weights)
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Fiber Distribution	Found in greatest abundance in muscles of the trunk, especially postural muscles	Found in greatest abundance in muscles of the lower limbs	Found in greatest abundance in muscles of the upper limbs																																														
Fiber Diameter	Smallest	Intermediate	Largest																																														
Number of Mitochondria	Many	Many	Few																																														
Amount of Myoglobin	Large	Large	Small																																														
Primary Fiber Function	Endurance (e.g., marathon running), maintaining posture	Medium duration, moderate movement (e.g., walking, biking)	Short duration, intense movement (e.g., sprinting, lifting weights)																																														
申覆	原因： 1. 最易疲乏的應該為(D) 2. 根據BRS cell biology and histology 第七版 第125頁表格，可知the most easily fatigued muscle為IIb，即D選項之fast anaerobic fibers，所以學生認為答案應該為D，謝謝老師 申覆結果：C→D 老師回覆：原答案誤植為C，修正為D																																																
出處	陳永佳 組織學-肌腱、韌帶、肌肉(ppt pg. 42)																																																

26	In resting muscle, which of the following proteins masks the myosin-binding site on the actin molecule? (A) myosin head (B) nebulin (C) tropomodulin (D) tropomyosin
答案	(D) tropomyosin

詳解	 ③ Calcium ions bind to troponin. Troponin changes shape, moving tropomyosin on the actin to expose active sites on actin molecules of thin filaments. Myosin heads of thick filaments attach to exposed active sites to form crossbridges. 由此可知在resting state，tropomyosin蓋住actin上面的myosin-binding site。
出處	陳永佳 組織學-肌腱、韌帶、肌肉(ppt pg. 20)

27	Which of the following statements is true when skeletal muscle contraction? (A) M line becomes shorter (B) H zone becomes narrower (C) A band becomes wider (D) I band becomes wider
答案	(B) H zone becomes narrower
詳解	骨骼肌收縮時： (A) M line是最中間的線-->不變。 (B) H Zone是放鬆時粗肌絲未與細肌絲重疊的部分-->收縮時變窄。 (C) A Band是粗肌絲的區域，即暗帶-->收縮時不變。 (D) I Band是細肌絲未與粗肌絲重疊的區域，即亮帶-->收縮時變短。

	Comparison of sarcomere in contracted and relaxed muscle (important!) Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display. Fill in the Blanks Which of the region will be 1. shortened 2. no changed 3. lengthened after muscle contraction? ROSS fig. 11.6 Sarcomeres in different functional stages.
出處	陳永佳 組織學-肌腱、韌帶、肌肉(ppt pg. 28)

28	Endochondral bone formation does NOT occur in the (A) femur (B) stapes (C) squamous part of temporal bone (D) greater wing of sphenoid bone
答案	(C) squamous part of temporal bone
詳解	Development of Cranium • From mesenchyme around the brain Box 9-2 Embryological Origins of Bones of the Cranium NEUROCRANIUM Chondrocranium Occipital Sphenoid Ethmoid Petrous and mastoid part of temporal Membranous neurocranium Interparietal part of occipital Parietal Frontal Squamous part of temporal PHARYNGEAL ARCH I Cartilaginous viscerocranium Meckel's cartilage Malleus Incus Membranous viscerocranium Maxillary process (superficial) Squamous part of temporal Zygomatic Maxillary Premaxillary Nasal? Lacrimal? Maxillary process (deep) Palatine Vomer Pterygoid laminae Mandibular process Mandible Tympanic ring PHARYNGEAL ARCH II Cartilaginous viscerocranium Reichert's cartilage Stapes Styloid process Carlson Human Embryology and Developmental Biology (c)petrous and mastoid part of temporal bone 才是
出處	何宛怡 肌肉骨骼胚胎學(ppt pg. 16)

29	Which of the following statement is TRUE concerning the development of the limb? (A) the zone of polarizing activity establishes the proximal-distal axis of the bud (B) the upper limbs rotate medially on their longitudinal axes during limb development (C) all of the secondary ossification centers appear in the limb after birth (D) each limb bud consists of a mesenchymal core of somatic mesoderm covered by surface ectoderm
答案	(D) each limb bud consists of a mesenchymal core of somatic mesoderm covered by surface ectoderm
詳解	(A) AER activity establishes the proximal-distal axis of the bud (B) the upper limbs rotate laterally on their longitudinal axes during limb development (C) all of the secondary ossification centers appear in the limb after birth (femur distal end/ tibial proximal end are the first to appear, present at birth)
申覆	原因： 在老師ppt pg.31提到 dermomyotome 會migrate 到 limb bud ，想詢問老師是否代表D選項在 surface ectoderm 與 somatic mesoderm 之間還會有一層 dermomyotome，因此D選項也錯誤？謝謝老師解答！ 申覆結果： 維持原答案 老師回覆： 因為dermomyotome只會出現在軀幹，而這一題是問limb bud，所以在ectoderm跟somatic mesoderm間不會有dermomyotome。答案不變。
出處	何宛怡 肌肉骨骼胚胎學(ppt pg. 27)

30	Which of the following structures is NOT derived from the hypaxial divisions of myotomes? (A) longissimus capitis (B) geniohyoid (C) rectus abdominis (D) levator ani
答案	(A) longissimus capitis
詳解	軸下肌群可分為四個部分，詳細如下圖所示。

詳解	(A)(B)理學檢查提到檢查肌肉力量與張力，需同時評估兩側是否對稱，因此不一定是對稱。其他例如自體免疫疾病中，也有以遠端肌肉無力的疾病。因此錯誤。 <i>Conote for KMU Medicine 109</i> 010 ◎ Physical examination 理學檢查 1. Muscle power (symmetric or asymmetric) 肌肉力量 (對稱或不對稱) 2. Muscle tone (floppiness) 肌肉張力 (鬆弛) 3. Deep tendon reflex 深層肌腱反射 (通常是 decrease) 4. Facial involvement (ptosis, ophthalmoplegia, high- arched palate) 臉部受侵犯 (眼瞼下垂、眼肌麻痺、上顎高拱) 5. Tongue fasciculation; hand tremor 舌肌顫搐；手顫抖 6. Joint contracture or hyperextensibility 關節攣縮或過度伸展 7. Skin rash (Gottron's papules and heliotrope); keloid 皮疹 (Gottron 氏丘疹和向陽疹)；蟹足腫 8. Spine/chest deformity; winged scapula 脊柱/胸部畸形；翼狀肩胛骨 9. Muscle hypertrophy or atrophy 肌肉肥大或萎縮 10. Trunk, tongue weakness 軀幹、舌頭無力
出處	梁文貞 肌肉疾病(M109共筆pg. 7, 10)

32	As for the physical findings of the patients with neuromuscular diseases, which description is correct? (A) Tongue fasciculation could be a characteristic of lower motor neuron disease. (B) Head lague and frog posture indicate increased muscle tone. (C) Joint contracture always develops in late non-ambulant stage (D) All above descriptions are correct.
答案	(A) Tongue fasciculation could be a characteristic of lower motor neuron disease.
詳解	(A) Correct <i>Conote for KMU Medicine 109</i> 013 五、 Lower motor neuron disease Tongue fasciculation  Character of lower motor neuron disease 1. (上課影片 28:24) Tongue fasciculation：可以看到舌頭一直在抖，這是病人不由自主的抖，無法控制。 2. Spinal muscular atrophy 脊髓肌肉萎縮症的病人就很常出現這種 Tongue fasciculation 的表現。 (B) Head lag and frog posture → Hypotonia

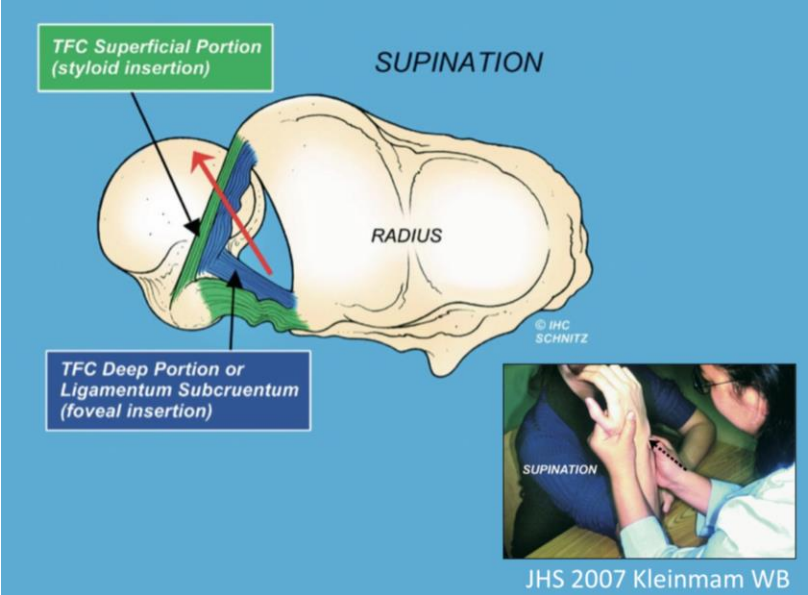
	三、 Frog posture 黃琳雅 編撰 即便只是讓嬰兒躺著，看起來都不一樣 (1) (左圖) Hypotonia (肌張力低下) 的嬰兒：呈現 Frog posture，四肢完全貼在床上。因為是 Severe Hypotonia，所以也伴隨著 weakness 的問題。 (2) (右圖) Normal muscle tone 的嬰兒：四肢 flexion，膝蓋都會離開床，不會貼著。  Courtesy of Prof Nonaka Frog posture
出處	梁文貞 肌肉疾病(M109共筆pg. 11, 13)

33	As for the feasible therapeutic strategy of neuromuscular diseases, which is correct? (A) nutrient (Vitamins and non-vitamin) supplementation (B) gene therapy (C) neuromuscular junction transmission modulation (D) All above are correction
答案	(D) All above are correction
詳解	Treatable NMD • Supplementation - Nutritional - Primary carnitine deficiency (carnitine) - Multiple acyl-CoA dehydrogenase deficiency (riboflavin) - Brown-Vialetto-Van Laere syndrome/Fazio-Londe syndrome (riboflavin) - Primary CoQ10 deficiency (CoQ10) - Thymidine kinase 2 deficiency (dC+dT) • Immune modulation - Autoimmune myositis - Myasthenic gravis - Immune-mediated neuropathy - Sporadic late-onset nemaline myopathy • NMJ transmission modulation - Myasthenic gravis - Congenital myasthenic syndrome - Lambert-Eaton myasthenic syndrome • Enzyme replacement - Pompe disease - Fabry disease • Gene therapy - Spinal muscular atrophy ✓ - Duchenne muscular dystrophy ✓ - X-linked myotubular myopathy - Beta-sarcoglycanopathy - Pompe disease - Some types of limb-girdle muscular dystrophy • Oligonucleotides (mRNA-targeting) - Duchenne muscular dystrophy - Spinal muscular atrophy - Amyloidosis 要診斷對 ★ 吃了就會好
出處	梁文貞 肌肉疾病(M109共筆pg. 33)老師上課講義ppt沒有

34	Which is correct about finger structure? (A) Proper digital artery is lying below (volar to) its corresponding digital nerve. (B) Triangular ligament prevents volar subluxation of the conjoint lateral bands with PIP joint flexion. (C) The function of Dorsal interossei is index and ring finger adduction. (D) Boutonniere deformity is caused by terminal tendon rupture.
答案	(B) Triangular ligament prevents volar subluxation of the conjoint lateral bands with PIP joint flexion.
詳解	(A) volar 改成 dorsal (C) adduction 改成 abduction (D) terminal tendon 改成 triangular tendon (central band/ slip)
出處	邱兆澤 手部、腕部骨科功能解剖學(ppt pg. 16)

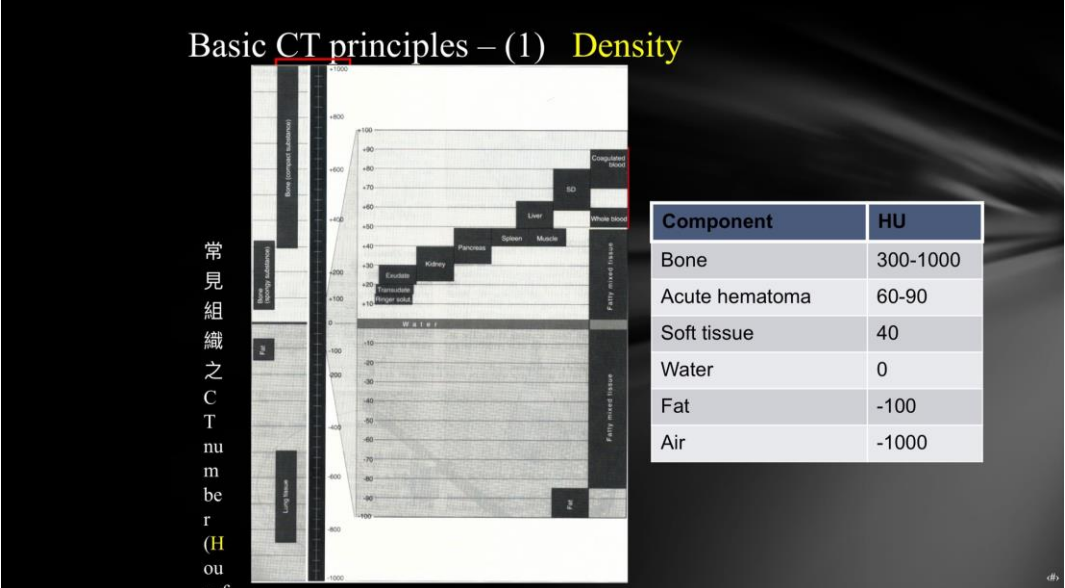
35	Which is not the extrinsic ligament of wrist? (A) Short radiolunate ligament (SRL) (B) Deltoid ligament (C) Radioscapophcapitate (RSC) ligament (D) Dorsal radiocarpal (DRC) ligament
答案	(B) Deltoid ligament
詳解	所謂的extrinsic的意思是要連接到radial bone或ulna bone，ACD都有看到中間有radio的字根，因此都不對，正確答案因此為B
出處	邱兆澤 手部、腕部骨科功能解剖學

36	Which is correct about wrist anatomic structure? (A) Volar deep fiber of Triangular fibrocartilage complex (TFCC) is lax while wrist pronation. (B) volar superficial fiber of Triangular fibrocartilage complex (TFCC) is lax while wrist supination. (C) Superficial head of Pronator quadratus plays major role of distal radioulnar joint stability. (D) TFCC superficial fiber inserts directly into the bony ulna styloid.
答案	(D) TFCC superficial fiber inserts directly into the bony ulna styloid.

詳解	 TFC Superficial Portion (styloid insertion) TFC Deep Portion or Ligamentum Subcruentum (foveal insertion) RADIUS SUPINATION © JHC SCHNITZ JHS 2007 Kleinmam WB
出處	邱兆澤 手部、腕部骨科功能解剖學(ppt pg. 43)

37	In the computed tomography image, which of the following values is closest to the density of fat? (A) 800 HU (B) 80 HU (C) 0 HU (D) -100 HU
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答案	(D) -100 HU
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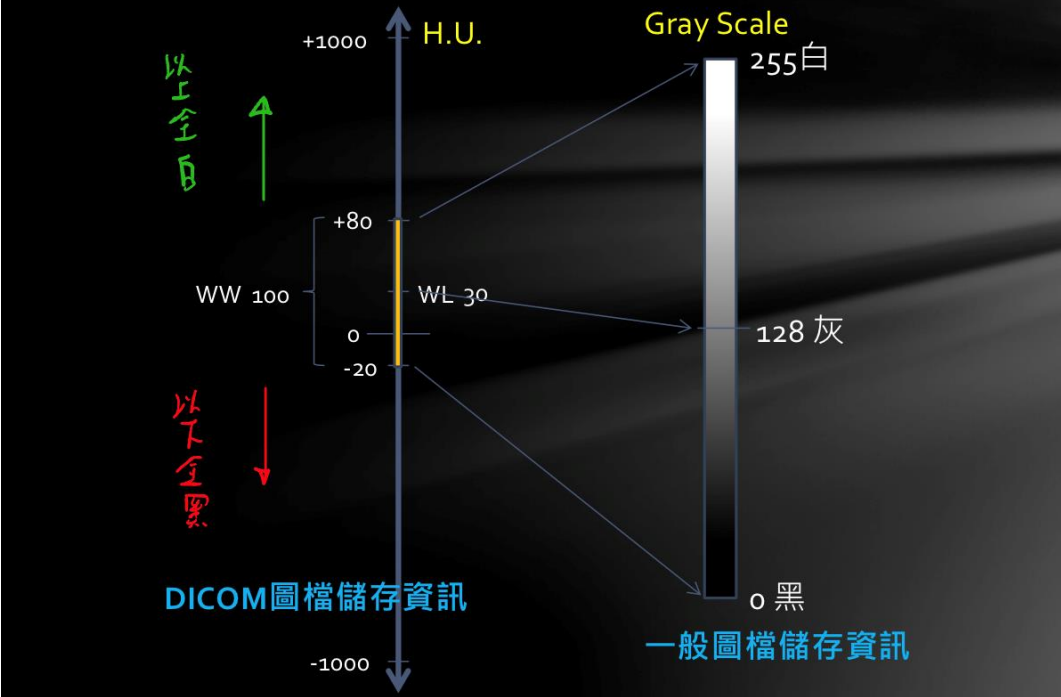
詳解	Basic CT principles – (1) Density  常見組織之CT number (HU) <table border="1"> <thead> <tr> <th>Component</th> <th>HU</th> </tr> </thead> <tbody> <tr> <td>Bone</td> <td>300-1000</td> </tr> <tr> <td>Acute hematoma</td> <td>60-90</td> </tr> <tr> <td>Soft tissue</td> <td>40</td> </tr> <tr> <td>Water</td> <td>0</td> </tr> <tr> <td>Fat</td> <td>-100</td> </tr> <tr> <td>Air</td> <td>-1000</td> </tr> </tbody> </table>	Component	HU	Bone	300-1000	Acute hematoma	60-90	Soft tissue	40	Water	0	Fat	-100	Air	-1000
Component	HU														
Bone	300-1000														
Acute hematoma	60-90														
Soft tissue	40														
Water	0														
Fat	-100														
Air	-1000														

右邊的圖表中可以知道fat就是-100HU

出處	蔡欣學 基礎影像學原理(ppt pg. 13)
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38	Which of the following sequence images are similar to the STIR sequence images in MRI examination? (A) T1WI + water suppression (B) T1WI + fat suppression (C) T2WI + water suppression (D) T2WI + fat suppression						
答案	(D) T2WI + fat suppression						
詳解	FLAIR $\approx$ T2WI + water suppression, Water- hypointensity, Edema- hyperintensity STIR $\approx$ T2WI + fat suppression (FS), Water- hyperintensity, Edema- hyperintensity MRI判讀前準備工作 Know common sequences Basic Advanced T2WI: Water- hyperintensity T1WI: Water- hypointensity FLAIR = T2WI + water suppression Water- hypointensity Edema- hyperintensity STIR = T2WI + fat suppression (FS) Water- hyperintensity Edema- hyperintensity T1WI + FS T2WI + FS T2WI + FS 亦可使用STIR <table border="1"> <thead> <tr> <th>T1 白質白</th> <th>T2 白質黑</th> </tr> </thead> <tbody> <tr> <td colspan="2">Fat : T1 and T2 都是亮 (high signal)的</td> </tr> <tr> <td colspan="2">Water : T2 亮 T1 暗</td> </tr> </tbody> </table>	T1 白質白	T2 白質黑	Fat : T1 and T2 都是亮 (high signal)的		Water : T2 亮 T1 暗	
T1 白質白	T2 白質黑						
Fat : T1 and T2 都是亮 (high signal)的							
Water : T2 亮 T1 暗							
出處	蔡欣學 基礎影像學原理(ppt pg. 21)						

39	In the image settings of computed tomography, if the window width is 100 and the window level is 0, what color does the water appear on the grayscale image? (A) white (B) gray (C) black (D) red
答案	(B) gray

詳解	Basic CT principles – (2) Window Width (WW) and Window Level (WL) 以brain常用之WW=100及WL=30設定為例  題目設定以0為基準線，寬度100表示上下各50單位，所以+50以上全白、-50以下全黑，水的HU值為0介於兩者之間，呈現灰色
出處	蔡欣學 基礎影像學原理(ppt pg. 14)

40	Which is correct about antineutrophil cytoplasmic antibodies (ANCA)? (A) PR3-ANCA is pANCA, MPO-ANCA is cANCA (B) In granulomatosis with polyangiitis, PR3-ANCA is more common than MPO-ANCA (C) Most microscopic polyangiitis (MPA) patients don't have ANCA (D) all of above
答案	(B) In granulomatosis with polyangiitis, PR3-ANCA is more common than MPO-ANCA
詳解	(A) PR3-ANCA (c-ANCA) ; MPO-ANCA (p-ANCA)

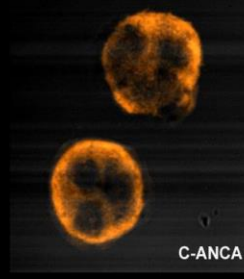
ANCA

- **PR3-ANCA; cANCA**
- **MPO-ANCA; pANCA**
- Atypical ANCA



P-ANCA

Ag: myeloperoxidase (MPO)



C-ANCA

Ag: proteinase 3 (PR3)

(B) 正確，口訣：GPA don't want 3.0 or C grade
記GPA 跟 PR3-ANCA =c-ANCA是一組的

ANCA

- **PR3-ANCA : 80-90% of GPA**
 - 20-40% of MPA, 20-40% of pauci-immune crescentic glomerulonephritis and 35% of EGPA
- **MPO-ANCA : 50% of MPA, 50% of primary pauci-immune necrotizing crescentic glomerulonephritis, 35% of EGPA**
- Atypical ANCA is associated with drug-induced systemic vasculitis, inflammatory bowel disease and rheumatoid arthritis

(C) MPO-ANCA 在 MPA 病人上較常見，佔50%

出處

曾家駿 免疫風溼學實驗室檢查(ppt pg. 30, 32)

41	Which combination of autoantibodies and clinical manifestation is correct? (A) HLA-B27 and thrombosis (B) Anti-β2GP1 and ankylosing spondylitis (C) Anti-histone and Drug-induced lupus (D) All of above is correct						
答案	(C) Anti-histone and Drug-induced lupus						
詳解	◎ HLA-B27 人類白血球抗原 B27 型 >90% of ankylosing spondylitis (僵直性脊椎炎) (國考) - Also related with other SpA (Spondyloarthritis) <i>雖然沒向上看那麼強</i> 在僵直性脊椎炎病患中，有超過 90% 的病人血液中測的到 HLA-B27 抗原。 也可能出現在其他僵直性脊椎炎的 subtype，例如： - Juvenile SpA 幼年型 - Undifferentiated SpA 未分化型 - Psoriatic arthritis 乾癬性關節炎 - Acute anterior uveitis 急性前葡萄膜炎 - Reactive arthritis 反應性關節炎 - Arthritis associated with Ulcerative Colitis/ Crohn's Disease 與潰瘍性結腸炎/克隆氏症相關的關節炎 - 但 HLA-B27 仍以 ankylosing spondylitis 表現為主 - 臨床意義：遇到有僵直性脊椎炎症狀的患者，要抽 HLA-B27 檢查；但只憑 HLA-B27 陽性不足以排除其他疾病，須配合臨床症狀與其他檢驗 1. Diffuse 瀰漫型、均質型 <table border="1"> <thead> <tr> <th>Antigen identified</th><th>Clinical correlation</th></tr> </thead> <tbody> <tr> <td>Deoxyribonucleoprotein</td><td>Non-specific</td></tr> <tr> <td>Histones</td><td>Drug-induced lupus</td></tr> </tbody> </table> - Deoxyribonucleoprotein 抗體會是 non-specific <i>☆</i> Histones 抗體會發生在 drug-induced lupus 的病人 (要記) (國考)	Antigen identified	Clinical correlation	Deoxyribonucleoprotein	Non-specific	Histones	Drug-induced lupus
Antigen identified	Clinical correlation						
Deoxyribonucleoprotein	Non-specific						
Histones	Drug-induced lupus						

◎ Antiphospholipid antibodies (APA) 抗磷脂質抗體

一、Antiphospholipid syndrome 抗磷脂症候群

1. 以下兩種狀況可以說病人有 Antiphospholipid syndrome

(1) venous or arterial thrombosis + APA

靜脈、動脈不正常的栓塞配合 APA 陽性

(2) Pregnancy morbidity + APA

容易流產配合 APA 陽性 (不同週數有不同次數流產定義)

二、Antiphospholipid antibodies (APA) 抗磷脂質抗體

1. APA 為一群抗體的總稱，非單一抗體 對抗 cell membrane Ab

2. Antiphospholipid antibodies include antibodies directed against phospholipid-associated proteins such as cardiolipin, β_2 -glycoprotein 1

APA 主要是針對和磷脂質相關蛋白質的抗體，比較有名的有 cardiolipin 的抗體、 β_2 -glycoprotein 1 的抗體。

因為 cardiolipin、 β_2 -glycoprotein 1 等抗原會在血管壁內皮細胞膜上產生，因此 APA 容易造成血管內皮細胞活化或造成栓塞。

3. APA 的種類

- ☆ {
- (1) Anticardiolipin antibody
 - (2) Lupus anticoagulant
 - (3) Anti- β_2 GP1 antibody

(4) 臨床上以上三種都會驗

(5) The overall positivity for any one of the APA studied in fetal loss was 42.6%

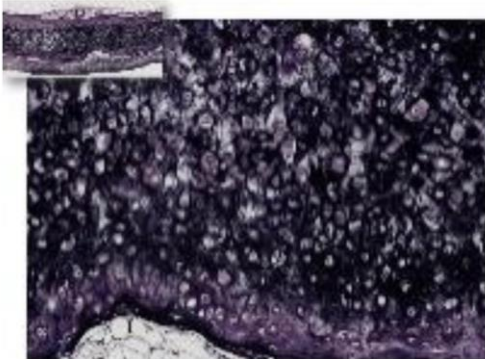
在這些病人中如果有流產，大概一半可以測到 APA

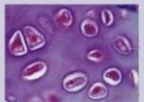
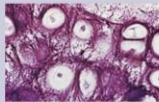
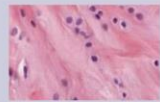
(不良流產、早產)

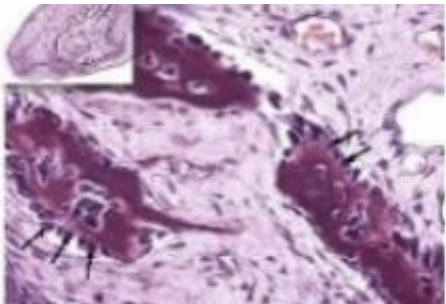
出處

曾家駿 免疫風溼學實驗室檢查(共筆pg. 11, 16, 19)

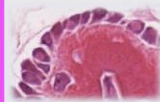
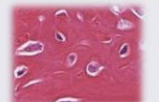
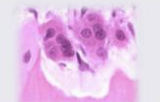
42	Mr. Chen, a 60 year-old patient with type 2 diabetes. He was admitted for Hyperosmolar Hyperglycemic State (HHS). During the second day of admission, sudden onset of right ankle swelling occurred, and the synovial fluid analysis by polarized microscopy was shown in the picture. What is the diagnosis?  (A) Septic arthritis (B) Gout (C) Calcium pyrophosphate deposition disease (D) Osteoarthritis
答案	(B) Gout
詳解	Needle shape, negative birefringent (針狀 負偏光)為 gout, rhomboid, positive birefringent (菱形 正偏光)為 pseudogout。
出處	曾家駿 免疫風溼學實驗室檢查

43	What kind of tissue is it?  (A) Hyaline cartilage (B) Elastic cartilage (C) Fibrocartilage (D) Bone
答案	(B) Elastic cartilage

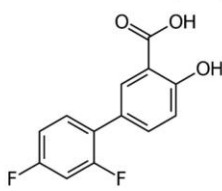
詳解		Hyaline cartilage	Elastic cartilage	Fibrocartilage
				
	Fiber	Type II collagen fibrils Aggrecan	Type II collagen fibrils Aggrecan Elastic fibers	Type II and type I collagen fibers Aggrecan Versican
	Lacunae	Large	Large	Small
	Perichondrium	Present	Present	None
	Main cell types	Chondrocytes	Chondrocytes	Chondrocytes Fibroblasts
出處	張瀛双 組織學與實驗-骨骼與軟骨			

44	What are the cell labeled by the arrows?  (A) Osteoblasts (B) Osteocytes (C) Osteoclasts (D) Chondrocytes
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答案	(A) Osteoblasts
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詳解		Osteoblast	Osteocyte	Osteoclast
				
				
	Location	Bone surface	Lacunae of bone matrix	Bone surface
	Precursor cells	Osteoprogenitors	Osteoblast	Hemopoietic cells
	Function	Deposits bone matrix Mineralization	Maintains bone matrix	Resorbs bone
Cell morphology		Cuboidal or polygonal Mononuclear Basophilic cytoplasm	Trapped within formed bone Mononuclear Basophilic cytoplasm	Large Multinuclear Acidophilic cytoplasm

出處	張瀛双 組織學與實驗-骨骼與軟骨
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45	All of the following statements regarding acetylated and nonacetylated salicylates are correct EXCEPT? (A) Diflunisal belongs to nonacetylated salicylates. (B) Diflunisal belongs to COX-1 selective inhibitor. (C) Aspirin selectively inhibits platelet function at low doses. (D) Aspirin reduces the incidence of transient ischemic attacks.
答案	(B) Diflunisal belongs to COX-1 selective inhibitor.
詳解	 Antipyretic Effects • ↓ elevated temp (inhibition of PGE_2), whereas normal body temp is only slightly affected Platelet Effects • slightly prolonged bleeding time, inhibition of platelet aggregation • low doses (60-80 mg/d): irreversibly inhibit TxA_2 production in platelets without markedly affecting production in the endothelial cells of the blood vessel (C)(D) (A) 六、Diflunisal (Dolobid®) 1. Difluorophenyl derivatives of salicylic acid Diflunisal 也是水楊酸的衍生物，但不會跟 Aspirin 一樣裂解成水楊酸 (1) (補)和 Aspirin 不同，此為非乙酰化(non-acetylated)的水楊酸類藥物 (2) (補)為水楊酸加上二氟苯基(Difluorophenyl)，結構圖在下方  老師的講義中並無提及Diflunisal為COX1 selective 根據其結構，並沒有乙酰基轉給COX酵素
出處	吳炳男 非類固醇抗發炎之藥理學

46	Which of the following statements regarding Acetaminophen is correct? (A) It has significant COX-2 inhibitory actions. (B) It has potent anti-inflammatory effects. (C) Dosing in adults is recommended not to exceed 10 g/d. (D) N-acetylcysteine can be used for those at risk of hepatic injury.							
答案	(D) N-acetylcysteine can be used for those at risk of hepatic injury.							
詳解	<table><tr><td rowspan="3">Acetaminophen (Scanol®、Panadol®、Depyretin®)</td><td>作用機轉</td><td>Acetaminophen 在☆中樞神經系統可以抑制 Prostaglandin 的合成，因此有解熱鎮痛的功用。其對周邊組織中的 COX enzyme 作用不大，故其只具微弱抗發炎活性。Acetaminophen 不會影響血小板功能或延長血液凝結時間。 ✓ 註：大部分是認為 Acetaminophen 只具解熱鎮痛，無抗發炎功效，故不屬於 NSAIDs，因為 NSAID 全名為 Nonsteroidal Anti-Inflammatory Drugs，Acetaminophen 不具抗發炎功效故不屬之。</td></tr><tr><td>藥動學</td><td>Acetaminophen 口服吸收快約 30-60 分鐘可達尖峰血中濃度，半衰期 2-3 小時，在正常劑量下，肝中之代謝物會和 Glutathione 結合成無毒的物質(Phase II 代謝)，服用過量的 Acetaminophen 會引起肝臟毒性，是因為毒性代謝物☆N-acetylimidoquinone 所造成，可以使用☆N-Acetylcysteine 來當作解毒劑(8 小時內投與最佳)。 ✓ 註：N-Acetylcysteine 就是常用的化痰劑，愛克痰(Acetin)的主成分。</td></tr><tr><td>治療用途</td><td>小兒科最常用的解熱鎮痛劑(因用 Aspirin 會增加 Reye syndrome 的危險性)。且 Acetaminophen 不會拮抗尿酸排除藥物 Probenecid 的作用，固可用於痛風患者。 ✓ Daily Acetaminophen maxium dose：☆4g /day。</td></tr></table>	Acetaminophen (Scanol®、Panadol®、Depyretin®)	作用機轉	Acetaminophen 在☆ 中樞神經系統 可以抑制 Prostaglandin 的合成，因此有解熱鎮痛的功用。其對周邊組織中的 COX enzyme 作用不大，故其只具微弱抗發炎活性。Acetaminophen 不會影響血小板功能或延長血液凝結時間。 ✓ 註：大部分是認為 Acetaminophen 只具解熱鎮痛，無抗發炎功效 ，故不屬於 NSAIDs，因為 NSAID 全名為 Nonsteroidal Anti-Inflammatory Drugs，Acetaminophen 不具抗發炎功效故 不屬之 。	藥動學	Acetaminophen 口服吸收快約 30-60 分鐘可達尖峰血中濃度，半衰期 2-3 小時，在正常劑量下，肝中之代謝物會和 Glutathione 結合成無毒的物質(Phase II 代謝)，服用過量的 Acetaminophen 會引起 肝臟毒性 ，是因為毒性代謝物☆ N-acetylimidoquinone 所造成，可以使用☆ N-Acetylcysteine 來當作解毒劑(8 小時內投與最佳)。 ✓ 註：N-Acetylcysteine 就是常用的化痰劑，愛克痰(Acetin)的主成分。	治療用途	小兒科最常用的解熱鎮痛劑 (因用 Aspirin 會增加 Reye syndrome 的危險性)。且 Acetaminophen 不會拮抗尿酸排除藥物 Probenecid 的作用，固可用於痛風患者。 ✓ Daily Acetaminophen maxium dose ：☆ 4g /day 。
Acetaminophen (Scanol®、Panadol®、Depyretin®)	作用機轉		Acetaminophen 在☆ 中樞神經系統 可以抑制 Prostaglandin 的合成，因此有解熱鎮痛的功用。其對周邊組織中的 COX enzyme 作用不大，故其只具微弱抗發炎活性。Acetaminophen 不會影響血小板功能或延長血液凝結時間。 ✓ 註：大部分是認為 Acetaminophen 只具解熱鎮痛，無抗發炎功效 ，故不屬於 NSAIDs，因為 NSAID 全名為 Nonsteroidal Anti-Inflammatory Drugs，Acetaminophen 不具抗發炎功效故 不屬之 。					
	藥動學		Acetaminophen 口服吸收快約 30-60 分鐘可達尖峰血中濃度，半衰期 2-3 小時，在正常劑量下，肝中之代謝物會和 Glutathione 結合成無毒的物質(Phase II 代謝)，服用過量的 Acetaminophen 會引起 肝臟毒性 ，是因為毒性代謝物☆ N-acetylimidoquinone 所造成，可以使用☆ N-Acetylcysteine 來當作解毒劑(8 小時內投與最佳)。 ✓ 註：N-Acetylcysteine 就是常用的化痰劑，愛克痰(Acetin)的主成分。					
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出處	吳炳男 非類固醇抗發炎之藥理學							

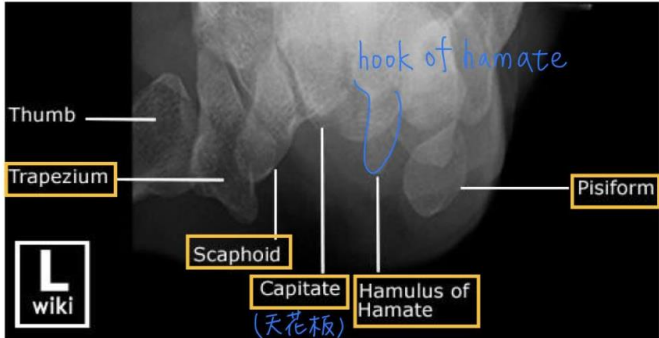
47	Which of the following NSAIDs is a prodrug and the only nonacid NSAID currently used? (A) Nabumetone (B) Meloxicam (C) Tenoxicam (D) Celecoxib		
答案	(A) Nabumetone		
詳解	◆Naphthylalkanones ➤ Nabumetone (Relifex®) • a weak COX inhibitor converted in the liver to its active metabolite, 6-methoxy-2-naphthylacetic acid (6-MNA) • selective inhibition of COX-2 • low incidence of GI side effect ◆Other selective COX-2 inhibitors ➤ Celecoxib (Celebrex®), Rofecoxib (Vioxx®) Etoricoxib (Arcoxia®), Valdecoxib (Bextra®) (A)上課時老師有提及唯一非酸性NSAID就是Nabumetone		
出處	吳炳男 非類固醇抗發炎之藥理學		

48	Which of the following statements regarding COX-2 inhibitors is correct? (A) There are no incidences of edema. (B) They have no impact on platelet aggregation at usual doses. (C) They increase circulating prostacyclin during long-term therapy. (D) They decrease the incidence of thrombotic risk.
答案	(B) They have no impact on platelet aggregation at usual doses.
詳解	只有COX-2 會影響 血小板 Fig. 1. A COX continuum? Two distinct genes for COX-1 and -2 may give rise to a number of constitutive and inducible COX proteins with overlapping functions. Considering prostanoid production by a COX continuum may help us appreciate which enzymes underlie prostanoid production in different tissues as well as the actions of traditional NSAIDs, newer COX-2-selective drugs, and acetaminophen ("COX-2b," as suggested in ref. 33; COX-3, as suggested in ref. 2). Warner TD & Mitchell JA: <i>PNAS</i>, 99: 13372, 2002.
出處	吳炳男 非類固醇抗發炎之藥理學(ppt pg. 5)

49	All of the following statements about anti-gout agents are correct except (A) Colchicine can reduce uric acid synthesis and excretion. (B) Probenecid and sulfinpyrazone are uricosuric agents. (C) Febuxostat reduces total uric acid body burden by inhibiting xanthine oxidase. (D) Indomethacin can be alternatively used in acute gout.
答案	(A) Colchicine can reduce uric acid synthesis and excretion.

詳解	➤ <u>Colchicine</u> • alkaloid, isolated from the autumn crocus (<i>Colchicum autumnale</i>) • binds to intracellular protein tubulin → preventing its polymerization into microtubules → inhibition of leukocyte migration & phagocytosis • blocks cell division by binding to mitotic spindles, also inhibits LTB₄ formation • minimal effects on uric acid synthesis & excretion • acute gouty arthritis • nausea, vomiting, abdominal pain, troublesome diarrhea Colchicine主要作用是抑制白血球的移動和吞噬作用，是針對急性痛風。
出處	吳炳男 非類固醇抗發炎之藥理學

50	Which of the following anti-gout agents is a recombinant mammalian urate oxidase that catalyzes the enzymatic oxidation of uric acid into allantoin, a more soluble and in active metabolite? (A) Colchicine (B) Allopurinol (C) Probenecid (D) Pegloticase
答案	(D) Pegloticase
詳解	➤ <u>Pegloticase (Krystexxa®)</u> • a recombinant mammalian uricase • converts uric acid to allantoin • recommended dose: 8 mg every 2 w (i.v. ≥ 2 h) • maintain low urate levels for up to 21 d • refractory chronic gout • anaphylaxis, infusion reaction, nephrolithiasis, arthralgia, muscle spasm, hemolytic anemia • should be avoided in pts with G6PD deficiency
出處	吳炳男 非類固醇抗發炎之藥理學

51	Which of the following is not the main structures in X-ray Wrist carpal tunnel view? (A) Hook of Hamate (B) Capitate (C) Metacarpal (D) Scaphoid
答案	(C) Metacarpal
詳解	(2) 手腕－2：Carpal Tunnel view (★可能考)  在carpal tunnel view 比較能看到的構造：capitate, scaphoid, trapezium, hamate, pisiform
出處	林育全 肌肉骨骼影像解剖學(1)-上肢

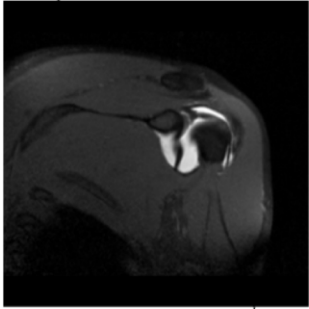
52	According to YCL method (CNRMT) interpreting a skeletal radiograph, which of the following belongs to Relation(R)? (A) A calcified lesion over greater tubercle in X-ray Shoulder AP view (B) Dislocated radius and ulna in X-ray Elbow lateral view (C) An over-bowing radius in X-ray Forearm AP view (D) Additional middle phalanx and distal phalanx in X-ray Hand AP view
答案	(B) Dislocated radius and ulna in X-ray Elbow lateral view

詳解	◎ 骨骼影像判讀 Interpret a Skeletal Radiograph 邱筱淇 編撰 一、YCL method (CNRMT) — 總論 Overview (YCL 是老師名字的縮寫，代表老師自己建立的判讀方法 XD) ★ 1. CNRMT 口訣 (會考)： (1) C: Category 看骨頭種類有沒有異常：有沒有多一種或少一種、有沒有出現異物。 (2) N: Number 看某種骨頭數量有沒有問題：例如—有人會有 6 根 metacarpals。 (3) R: Relation 看骨頭之間的相對位置有沒有問題。如： A. 排列(alignment)：是否有內翻或外翻、旋轉偏移角度(M106)。 B. 關節的空間(joint space)：是否有比較寬或窄。 (4) M: Morphology 看骨頭型態有沒有問題。如：形狀(shape)、尺寸(size)。 (5) T: Texture (lesion, density, fracture) 看材質 (質地) 有沒有問題。如： A. 骨頭裡面有病灶(lesion)，跟原本的骨頭看起來就不一樣。 (例如：原本海綿骨會有白色條紋，但因為有 lesion 導致條紋可能消失) B. 骨密度(density)：變高 (X 光上變較白)、變低 (X 光上變較黑) ② 骨密度較低，X 光穿透力就比較好，影像上看起來會比較黑。 C. 骨折(fracture)：骨頭裡面出現一條黑黑的骨折裂痕。
出處	林育全 肌肉骨骼影像解剖學(1)-上肢

53	According to YCL method (CNRMT) interpreting a skeletal radiograph, which of the following belongs to Morphology (M)? (A) A calcified lesion over greater tubercle in X-ray Shoulder AP view (B) Dislocated radius and ulna in X-ray Elbow lateral view (C) An over-bowing radius in X-ray Forearm AP view (D) Additional middle phalanx and distal phalanx in X-ray Hand AP view
答案	(C) An over-bowing radius in X-ray Forearm AP view
詳解	考古題

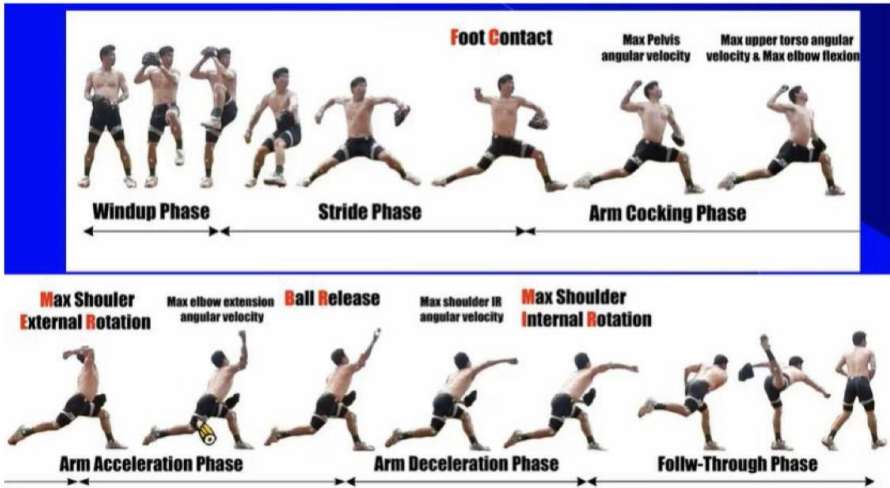
	◎ 骨骼影像判讀 Interpret a Skeletal Radiograph 邱筱淇 編撰 一、YCL method (CNRMT)—總論 Overview (YCL 是老師名字的縮寫，代表老師自己建立的判讀方法 XD) ★ 1. CNRMT 口訣 (會考)： (1) C: Category 看骨頭種類有沒有異常：有沒有多一種或少一種、有沒有出現異物。 (2) N: Number 看某種骨頭數量有沒有問題：例如—有人會有 6 根 metacarpals。 (3) R: Relation 看骨頭之間的相對位置有沒有問題。如： A. 排列(alignment)：是否有內翻或外翻、旋轉偏移角度(M106)。 B. 關節的空間(joint space)：是否有比較寬或窄。 (4) M: Morphology 看骨頭型態有沒有問題。如：形狀(shape)、尺寸(size)。 (5) T: Texture (lesion, density, fracture) 看材質(質地)有沒有問題。如： A. 骨頭裡面有病灶(lesion)，跟原本的骨頭看起來就不一樣。 (例如：原本海綿骨會有白色條紋，但因為有 lesion 導致條紋可能消失) B. 骨密度(density)：變高 (X 光上變較白)、變低 (X 光上變較黑) ② 骨密度較低，X 光穿透力就比較好，影像上看起來會比較黑。 C. 骨折(fracture)：骨頭裡面出現一條黑黑的骨折裂痕。
出處	林育全 肌肉骨骼影像解剖學(1)-上肢

54	Which statement(s) is/are false? (A) Brittle material undergoes very little plastic deformation before failure. (B) Tendon is a brittle material. (C) Ductile material undergoes more plastic deformation before failure. (D) Ceramics is a brittle material.
答案	(B) Tendon is a brittle material.
詳解	Tendon is a ductile Materials Brittle Materials Rupture occurs soon after elastic limit Plastic deformation is minimal e.g., PMMA Orthopedic Biomechanics Ductile Materials Undergo a large amount of plastic deformation prior to failure e.g., Tendon Orthopedic Biomechanics
出處	周伯禧 基礎生物力學

55	Which of the following(s) can be helpful in diagnosing Superior Labrum Anterior to Posterior Lesion (SLAP)? (A) MR-Arthrography (B) Injury Mechanism & Clinical S/S (C) O' Brien' s Test (D) All of the above
答案	(D) All of the above
詳解	SLAP Superior Labrum, Anterior to Posterior Lesions of the Shoulder --- Andrews JR, <i>Orthop Trans</i>, 1984. --- Snyder SJ, <i>Arthroscopy</i>, 1990. SLAP / History Pain with overhead activities --- time before scope : 10 - 29 months Mechanical symptoms of locking, catching, or grinding Impingement-type pain Dead arm syndrome Shoulder SLAP / Mechanism Traction Injuries (Maffet, 1995): --- sudden pull in an inferior direction --- throwing or overhead motion (Andrews, 1984) Compression Injuries (Snyder, 1990, 1998): --- fall on an outstretched arm --- slight forward flexion and abduction Direct blow to the shoulder (17%) Insidious onset with no trauma (22%) SLAP Physical Examination Speed's biceps tension test Compression -rotation test Hawkins test O'Brien test Jobe's relocation (Internal Impingement) Biceps load test II (Kim's) sensitivity: 89.7% ; specificity: 96.7% --- Kim SH, et al; <i>Arthroscopy</i>, 2001. Shoulder Diagnostic Imaging CT Arthrography MRI MR Arthrography MR Arthrography with arm traction / ext. rotation --- Chan KK, et al, <i>AJR</i>, 1999.  Shoulder

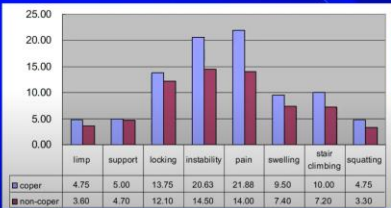
	1. 受傷機制&臨床症狀 2. PE (O' Brien test) 以及 3. 影像學的MR Arthrography 都對於SLAP的診斷有所幫助。
出處	周伯禧 基礎生物力學

56	Motion analysis can be clinically applied in which of the followings? (A) Study of baseball pitching (B) Simulation of fall on the outstretched hand (C) Gait Analysis in patients with ACL instability (D) All of the above
答案	(D) All off the above
詳解	Motion analysis定義廣泛, 關於Kinesiology等等都可以算其中一部分 其可分為Kinematics和Kinetic Kinematics是study of motion, 指動作上的差異, 觀察物體如何運動 Kinetics是計算關節在運動中受力的大小 所以(A)(B)(C)都包含在以上兩者的定義中
出處	周伯禧 基礎生物力學

57	The highest shoulder joint loading can occur on which of the time interval during pitching motion? (A) Winding phase (B) Late cocking & early acceleration phase (C) Deceleration phase (D) Follow through phase
答案	(B) Late cocking & early acceleration phase
詳解	

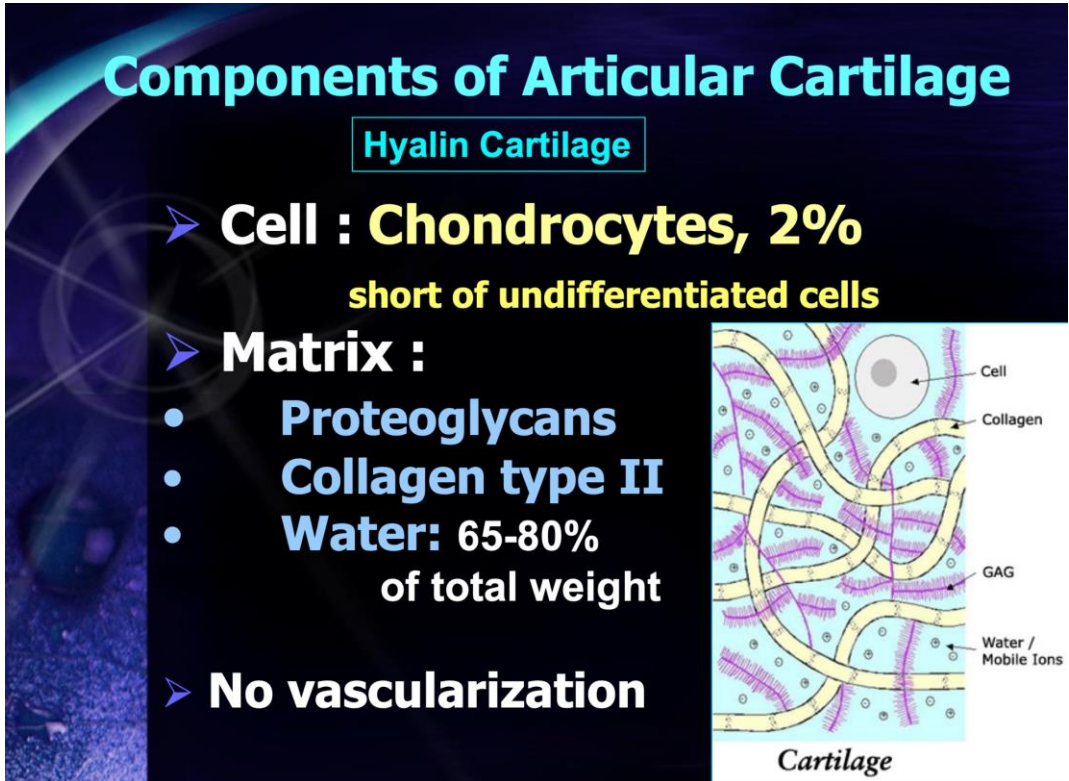
	在Late cocking & early acceleration phase時期，肩膀和手肘承受最大的力量，所以這時候肩膀與肘關節最容易受傷。
出處	周伯禧 運動創傷及運動醫學

58	Which of the following statement(s) about PCL (Posterior Cruciate Ligament) injuries is/are false (A) related to trauma and traffic accident (B) PCL reconstruction is the best treatment for isolated PCL tear (C) Posterior Sag & Posterior Drawer Test can detect PCL injuries (D) MRI is useful in diagnosing PCL injuries
答案	(B) PCL reconstruction is the best treatment for isolated PCL tear

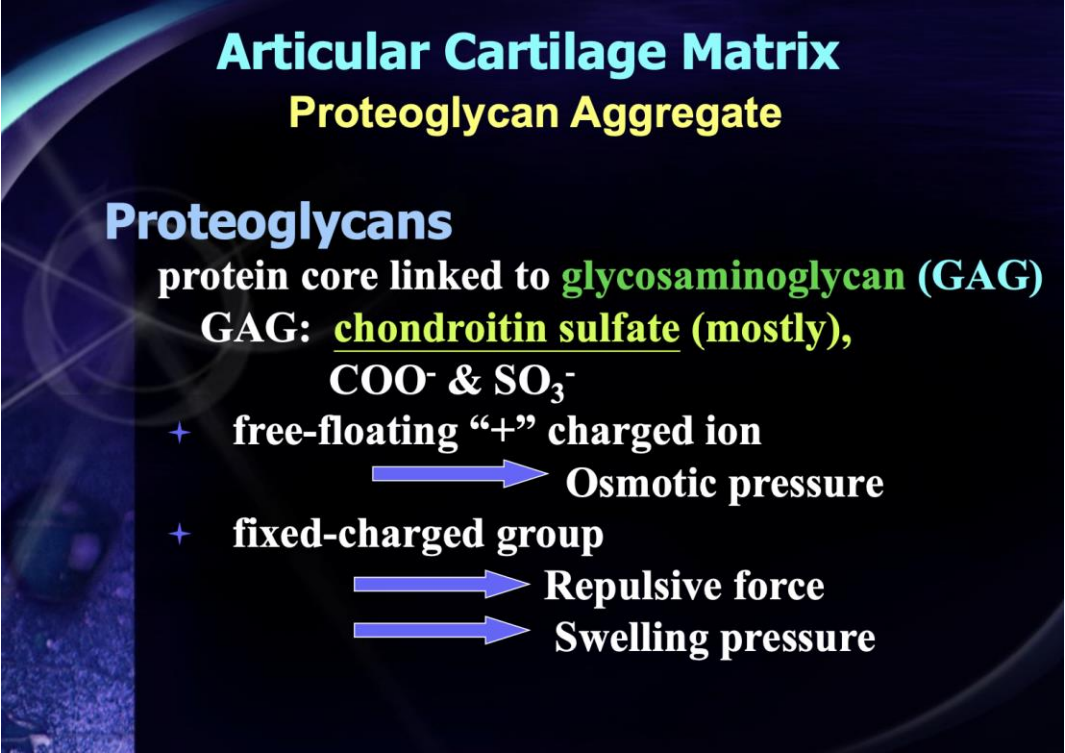
詳解	老師口述：單一PCL斷裂但復原良好組(Coper組)，找20位經過積極復健3個月以上的病患來進行測試，復原良好組會有比較好的肌肉強度，也會利用力量轉移的方式跟比較小的膝蓋角度來減輕膝蓋的負擔，不一定要開刀。 20 patients with isolated PCL rupture Failure from > 3 months of conservative treatment; unsatisfactory ADL, pain in injury limb. Coper 7M 3F Non-coper 8M 2F Control 6M 4F Ascending stairways ➤ There were no significant group and side to side differences in either kinematics or kinetics variables. <table><tr><th></th><th>limp</th><th>support</th><th>locking</th><th>instability</th><th>pain</th><th>swelling</th><th>stair climbing</th><th>squatting</th></tr><tr><td>□ coper</td><td>4.75</td><td>5.00</td><td>13.75</td><td>20.63</td><td>21.88</td><td>9.50</td><td>10.00</td><td>4.75</td></tr><tr><td>■ non-coper</td><td>3.60</td><td>4.70</td><td>12.10</td><td>14.90</td><td>14.00</td><td>7.40</td><td>7.20</td><td>3.30</td></tr></table> Conclusion □ Maintaining good muscle strength are one of adaptation strategies of copers. □ Keeping less knee flexion and tend to shift the loading to hip and ankle joint are useful strategies to decrease the loading . □ A rehabilitation program can be designed to strengthen the hip, knee extensors and the ankle flexors in order to help patients with PCL deficiency. 59		limp	support	locking	instability	pain	swelling	stair climbing	squatting	□ coper	4.75	5.00	13.75	20.63	21.88	9.50	10.00	4.75	■ non-coper	3.60	4.70	12.10	14.90	14.00	7.40	7.20	3.30
	limp	support	locking	instability	pain	swelling	stair climbing	squatting																				
□ coper	4.75	5.00	13.75	20.63	21.88	9.50	10.00	4.75																				
■ non-coper	3.60	4.70	12.10	14.90	14.00	7.40	7.20	3.30																				
出處	周伯禧 運動創傷及運動醫學																											

59	Valgus Extension Overload is probably the most common cause of elbow injury in base ball pitchers. It is not associated with (A) medial collateral ligament instability (B) osteochondral injuries of the radio-capitellar joint (C) radial nerve palsy (D) posterior elbow
答案	(C) radial nerve palsy
詳解	非常古，背誦題
出處	周伯禧 運動創傷及運動醫學

60	In normal bone growth process, where and when to process the endochondral ossification? Which one of the follows is the most appropriate answer? (A) At epiphyseal growth plate while long bone elongation (B) At epiphysis of a long bone while primary ossification (C) At center of a long bone while secondary ossification (D) At calvaria while fetal bone growth
答案	(A) At epiphyseal growth plate while long bone elongation
詳解	BC寫反，D頭蓋骨為膜內骨化 The diagram illustrates the four stages of endochondral ossification in a long bone: (e) 骨髓腔的形成 (Formation of the marrow cavity): Shows the initial cartilaginous model with a central marrow cavity. (f) 頭圓的加厚及增長 (Thickening and growth of the epiphysis): Shows the epiphysis becoming more prominent. (g) 次級骨化中心的形成 (Formation of the secondary ossification center): Shows the development of a secondary ossification center in the epiphysis. (h) 保留軟骨當做關節軟骨及骨骼板 (Retention of cartilage as articular cartilage and epiphyseal growth plate): Shows the final stage where articular cartilage and the epiphyseal growth plate are retained. Labels in the diagram include: Primary Ossification Center, Secondary Ossification Center, Articular cartilage, Epiphyseal growth plate, 骨髓腔 (Marrow cavity), 次級骨化中心 (Secondary ossification center), 關節軟骨 (Articular cartilage), and 骨骼板 (Epiphyseal growth plate).
出處	何美玲 骨與軟骨生理學

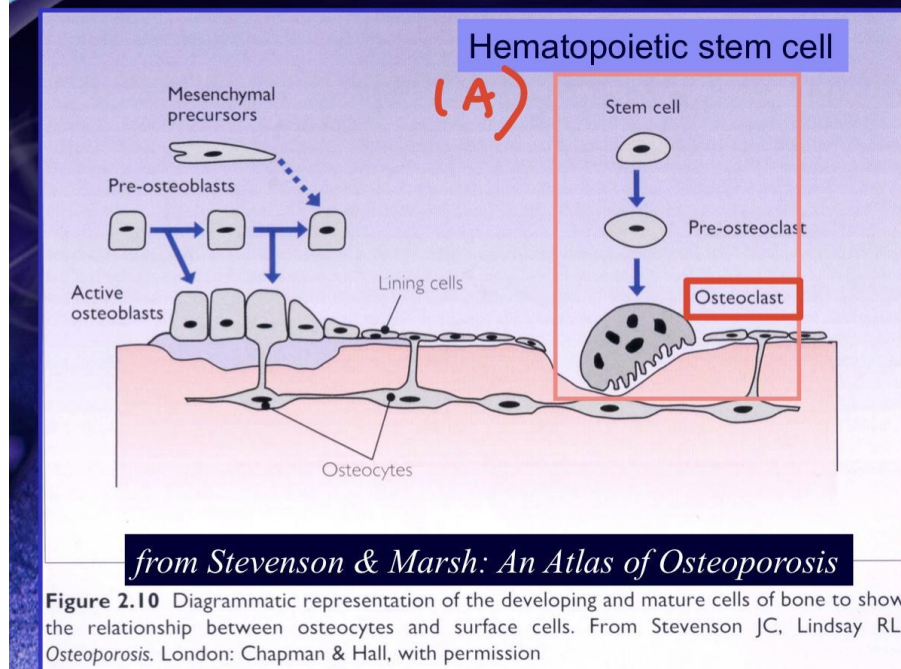
61	Which one of the follows is the most inappropriate for chondrocyte' s characteristics? (A) Chondrocytes are the cells in cartilages. (B) The fate of the chondrocytes in epiphyseal growth plates are undergoing terminal differentiation eventually while bone growth. (C) The chondrocytes in articular cartilage secrete collagen type I and glycosamino-glycan to maintain the normal hyalin cartilage. (D) The chondrocytes in normal articular cartilage does not undergo terminal differentiation.
答案	(C) The chondrocytes in articular cartilage secrete collagen type I and glycosamino-glycan to maintain the normal hyalin cartilage.
詳解	(C)是分泌collagen type II 
出處	何美玲 骨與軟骨生理學(ppt pg. 18)

62	Which one of the following statements is the most appropriate for articular cartilage? (A) Articular cartilage gets the nutrient from circulation by Haversian' s system. (B) Glycosaminoglycan with the negative charges on its molecular surface, which is helpful for influx of synovial fluid. (C) Collagen type II with triple helix structure, which is for maintaining the fluid in car tilage. (D) Collagen type I with triple helix structure, which is for maintaining tensile property in cartilage.
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答案	(B) Glycosaminoglycan with the negative charges on its molecular surface, which is helpful for influx of synovial fluid.
詳解	GAG: Chondroitin sulfate (mostly) A. 上面這個用紅圈圈起來的部分主要成分是 COO^- 和 SO_3^-，他們 因為負電相斥會造成 repulsive force，進而產生 swelling pressure 整個就會膨脹起來 B. Free-floating “+” charged ion $\rightarrow$ osmotic pressure 而當膨脹之後由於整個結構帶負電，正離子和水就會從外面的 synovial fluid 被吸進來進行營養交換進而達到保濕和代謝的功用  Articular Cartilage Matrix Proteoglycan Aggregate Proteoglycans protein core linked to glycosaminoglycan (GAG) GAG: <u>chondroitin sulfate (mostly)</u>, COO^- & SO_3^- + free-floating “+” charged ion $\rightarrow$ Osmotic pressure + fixed-charged group $\rightarrow$ Repulsive force $\rightarrow$ Swelling pressure
出處	何美玲 骨與軟骨生理學(ppt pg. 23)

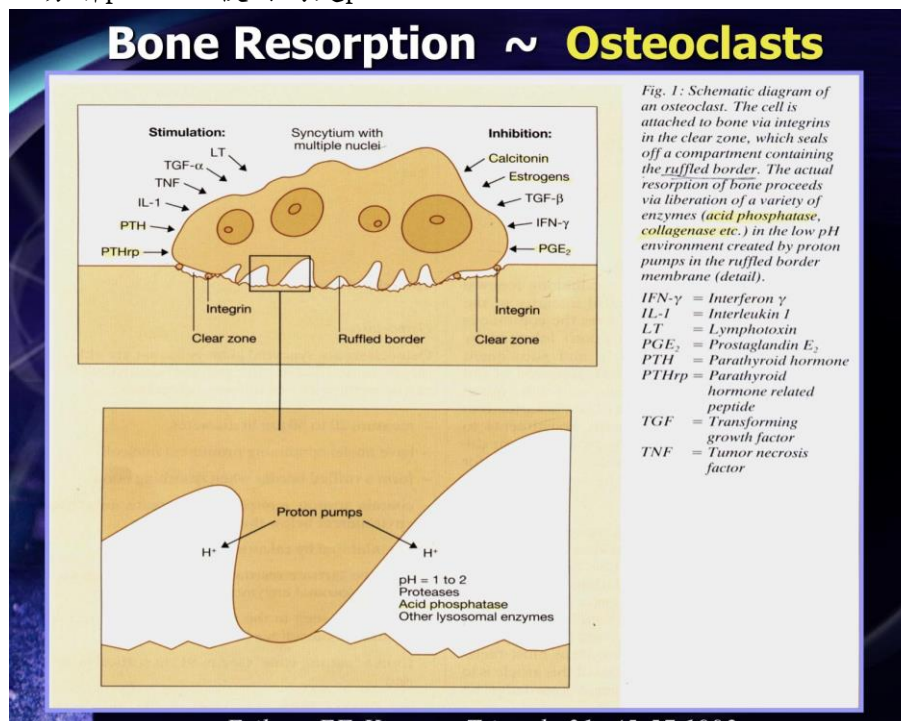
63	Which of the follows is the most appropriate for osteoclasts? (A) Osteoclasts are differentiated from bone marrow mesenchymal stem cells. (B) Osteoclasts are the most important bone cells to do bone formation (C) Osteoclasts are multi-nuclear macrophage-like cells, majorly for bone resorption. (D) Osteoclasts with proton pump, which is to digest bone matrix protein.
答案	(C) Osteoclasts are multi-nuclear macrophage-like cells, majorly for bone resorption.
詳解	(A) Osteoclasts 分化來自Hematopoietic stem cell.

The mature osteoclasts & bone resorption



(B) Osteoclasts對於bone resorption最為重要

(D) Osteoclasts的proton pump所創造的pH1-2的環境是為了分解磷酸鈣；欲分解protein使用的是protease.

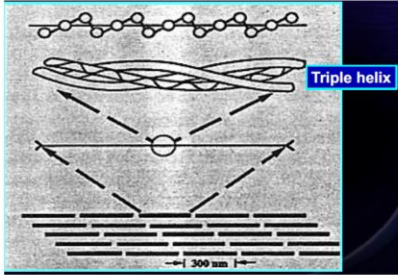


出處

何美玲 骨與軟骨生理學

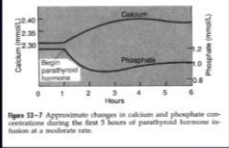
64	Which of follows is the most inappropriate for Osteoblasts? (A) Osteoblasts are differentiated from bone marrow mesenchymal stem cells. (B) Some of the osteoblasts differentiate into lining cells located on bone surface. (C) Most of the osteoblasts differentiate into osteocytes. (D) Osteocytes are the most important cells for bone formation.
答案	(D) Osteocytes are the most important cells for bone formation.
詳解	(D) Bone formation和Bone resorption都是在osteoblast和osteoclast相互調控平衡下進行， Bone formation又更需要osteoblast製造基質，相較之下osteocyte的角色就沒那麼重要。
申覆	原因： 根據Robins組織課本，osteoblasts僅10%~20%形成osteocytes，其他大部分會apoptosis。故C選項亦不妥。 申覆結果： 維持原答案 老師回覆： 雖然osteoblasts有很大的部份是apoptosis，但我們在這裡是強調osteoblasts繼續分化的細胞主要是osteocytes。 本題之命題是最不適宜的答案還是D
出處	何美玲 骨與軟骨生理學

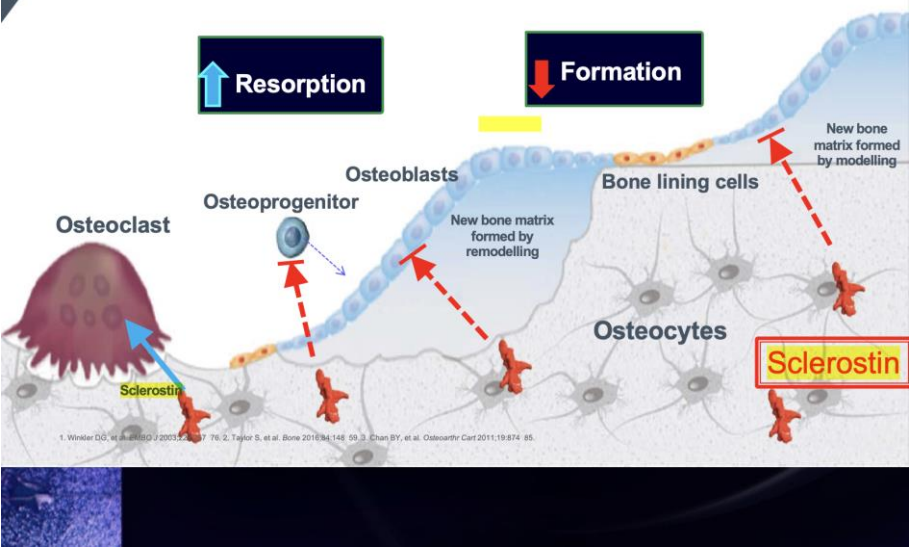
65	Which one of the follows is the most appropriate for keeping bone' s hardness (stiffness) to resist compression force? (A) Hydroxyapatite (B) Osteocalcin (C) Collagen type (D) Osteopontin
答案	(A) Hydroxyapatite

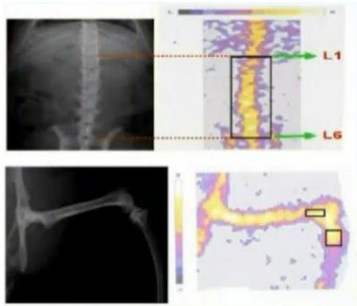
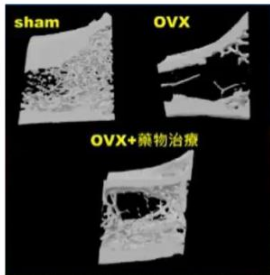
詳解	八、Bone Formation 骨生成 - Cells：Osteoblasts 成熟後變成 Osteocytes (1) 剛才才提到 Osteoblast 成熟後一部分埋在骨基質內形成 Osteocyte，一部分留在表面成為 Lining cell - Bone matrix：骨基質中軟的部分 (1) Collagen type 1：增加骨頭的張力，使組織較有彈性，不會一摔就碎  A. 身體其他組織如皮膚也有 collagen type 1 的存在 B. 上圖說明 Collagen type 1 跟 Collagen type 2（在軟骨中）一樣屬於長鏈的 triple helix，但成分不同 - Mineral：骨基質中硬的部分 (1) 主要由特殊結構的磷酸鈣組成，稱為 Hydroxyapatite，化學式為 $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$，結構十分穩固，臨床上作為骨粉能填充骨頭中缺損的部分 骨質中堅硬的部分主要由特殊結構的磷酸鈣組成稱為hydroxyapatite
申覆	原因： 根據Robbin's pathology，有一種跟type I collagen有關的疾病為osteogenesis imperfecta，課本對其的描述為：Osteogenesis imperfecta (OI), or brittle bone disease, the most common inherited disorder of connective tissue, is a phenotypically heterogenous disorder caused by deficiencies in type I collagen synthesis. ... The fundamental abnormality in OI is too little bone, resulting in extreme skeletal fragility. 顯示該疾病主因是type I collagen表現缺失造成骨頭異常脆弱，反過來說可以支持type I collagen對於骨頭的應力抵抗是極為重要的，因此老師您的題目問及stiffness的話，type I collagen我認為也是合理的答案。縱使這是出自病理學教科書，但我認為知識無遠弗屆，對於骨頭總是有一套放諸四海皆準的道理，望老師考慮將C也開放給分，非常謝謝老師 申覆結果：維持原答案 老師回覆： 在bone matrix 中hydroxyapatite提供抵擋compression force,而 collagen type I is for tensile force. 故最適宜答案是A. 當然若骨的成份出錯其微結構出問題，後續均都可能造成易損壞。
出處	何美玲 骨與軟骨生理學(共筆pg. 41)

66	Which one of the follows is the most appropriate for promoting osteoclastogenesis? (A) Estrogens (B) OPG (Decoy receptor of RANKL) (C) Calcitonin (D) RANKL
答案	(D) RANKL
詳解	RANKL與hematopoietic stem cell上的RANK結合後會促進其分化為osteoclast，反之OPG為RANKL的decoy receptor，會防止其結合至RANK而estrogen及calcitonin皆會抑制osteoclast的分化過程。 Osteoclastogenesis OPG/RANKL/RANK pathway RANK: Receptor for activator of nuclear factor-κB RANKL: Ligand of RANK OPG: Decoy receptor of RANKL from SCIENCE VOL 289, 2000
出處	何美玲 骨與軟骨生理學

67	Which one of follows is the most appropriate for the function of PTH (parathyroid hormone)? (A) Decrease circulating calcium level (B) Promote bone resorption physiologically (C) Elevate circulating phosphate level (D) Suppress bone formation at pharmacological dose
答案	(B) Promote bone resorption physiologically

詳解	Parathyroid Hormone (PTH) a polypeptide, 84 a.a., from chief cells Action • Plasma: $\uparrow \text{Ca}^{+2}$, $\downarrow \text{PO}_4^{-3}$ • Bone : $\uparrow$ bone resorption @ longer time scale : $\uparrow$ osteoblast, $\uparrow\uparrow$ osteoclast activity $\Rightarrow$ bone resorption, formation time and dose concern • Kidney : $\downarrow$ reabsorption of PO_4^{-3} $\uparrow$ reabsorption of Ca^{+2}  PTH生理功能為升血鈣降血磷，增加骨骼Resorption
出處	何美玲 骨與軟骨生理學(ppt pg. 60)

68	Which one of the following factors is to increase bone resorption but inhibit bone formation physiologically? (A) PTH (B) Calcitonin (C) Vitamine D3 (D) Sclerostin
答案	(D) Sclerostin
詳解	Sclerostin inhibits bone formation, but increases bone resorption  ↑ Resorption ↓ Formation Osteoclast Osteoprogenitor Osteoblasts Bone lining cells Osteocytes Sclerostin New bone matrix formed by remodelling New bone matrix formed by modelling 1. Winkler DG, et al. <i>Am J Pathol</i> 2003;161:177-76. 2. Taylor S, et al. <i>Bone</i> 2016;84:148-59. 3. Chan BY, et al. <i>Osteoarthritis Cartil</i> 2011;19:874-85.
出處	何美玲 骨與軟骨生理學(ppt pg. 46)

69	In an osteoporosis animal experiment, which one of the following techniques is the most appropriate to measure BV/TV (ratio of trabecular volume to total volume)? (A) Histomorphometry (B) Bone mineral density measurement (C) Micro-CT analysis (D) MRI analysis																								
答案	(C) Micro-CT analysis																								
詳解	二、骨質疏鬆症之評估 1. 雙光子骨質密度測量儀 DXA：骨密度 →測量 spine、femur、tibia 之骨密度。  2. (考古) Micro-CT：骨微結構 →可計算 tibia 的 bone volume(BV)和 total volume(TV)，以及 BV/TV 的百分比等之微結構變化，並可以呈現 3D 立體結構。 <table><tr><th colspan="2">Mirco-CT 常用分析項目</th><th>單位</th></tr><tr><td>Total volume</td><td>(TV)</td><td>mm³</td></tr><tr><td>Bone volume</td><td>(BV)</td><td>mm³</td></tr><tr><td>Percent bone volume</td><td>(BV/TV)</td><td>%</td></tr><tr><td>Bone surface / volume ratio</td><td>(BS/BV)</td><td>1/mm</td></tr><tr><td>Trabecular thickness</td><td>(Tb.Th)</td><td>mm</td></tr><tr><td>Trabecular number</td><td>(Tb.N)</td><td>1/mm</td></tr><tr><td>Trabecular separation</td><td>(Tb.Sp)</td><td>mm</td></tr></table> 	Mirco-CT 常用分析項目		單位	Total volume	(TV)	mm ³	Bone volume	(BV)	mm ³	Percent bone volume	(BV/TV)	%	Bone surface / volume ratio	(BS/BV)	1/mm	Trabecular thickness	(Tb.Th)	mm	Trabecular number	(Tb.N)	1/mm	Trabecular separation	(Tb.Sp)	mm
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出處	何美玲 生理學實驗-骨質疏鬆動物模式																								

70	Regarding the description of osteonecrosis of femoral head, which of the following is correct? (A) Drinking alcohol is the main cause of femoral head avascular necrosis in women (B) Crescent sign is a typical finding in X ray in stage III of femoral head osteonecrosis (C) HIV infection is a risk factor even without protease inhibitors (D) Donut-like change could be seen in MRI in avascular necrosis of the femoral head
答案	(C) HIV infection is a risk factor even without protease inhibitors

詳解

- (A) 老師上課特別強調，骨頭疾病多數女性發生率高，少數像是骨壞死反而是男性發生率高，其中男生常見的原因又以喝酒居多
- (B) Crescent sign是在stage II就可以看到的finding
- (C) 正確，HIV病毒的感染本身就是骨壞死的risk factor
- (D) donut like是核子醫學檢查的finding，MRI是double line sign

Epidemiology

- 10 percent of all total hip replacements (THR)
- Male > Female
- **Alcohol-associated ON: male**
- **Systemic lupus erythematosus (SLE)- associated ON: females**
- Usually less than 50 years
- **Bilateral hips involved 80% of the time**
- **3% of patients with osteonecrosis have multifocal involvement**

Risk Factors of Femoral Head ON

Direct Cause

- Irradiation
- Trauma
- Hematologic diseases (leukemia, lymphoma)
- Dysbaric disorders (Caisson disease)
- Marrow-replacing diseases (e.g. Gaucher's disease)
- Sickle cell disease

Indirect Cause

- Alcoholism
- Hypercoagulable states
- Steroids
- Systemic lupus erythematosus (SLE)
- Transplant patient
- Virus (CMV, hepatitis, HIV, rubella, rubeola, varicella)
- Protease inhibitors (type of HIV medication)
- Idiopathic

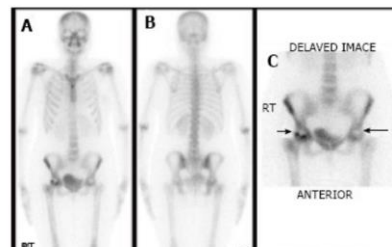
Diagnosis

3 Computed tomography scanning:

reveals zones of osteosclerosis surrounding the necrotic bone and repaired bone or shows subchondral bone fracture

4 Radionuclide examination:

- An acute-phase bone scan of the femoral head can reveal cold areas; scanning during the necrosis repair period demonstrates a cold area in the middle of a hot area, in other words a "donut-like" change (cold in hot).



Association Research Circulation Osseous (ARCO) international classification of osteonecrosis

Table 1

Association Research Circulation Osseous (ARCO) international classification of osteonecrosis [60].

ARCO stage	Image findings	Anteroposterior images	Description	3A (early) 3B (late)	Subchondral fracture on X-ray or CT		Subchondral fracture, fracture in the necrotic portion and/or flattening of the femoral head is seen on plain radiography or CT scan. →3A: Femoral head depression ≤ 2 mm →3B: Femoral head depression >2 mm
1	X-ray normal MRI abnormal		A band lesion of low signal intensity around the necrotic area is seen on MRI. A cold spot is seen on bone scan. No changes are seen on plain radiographs.				
2	X-ray abnormal MRI abnormal		Osteosclerosis, focal osteoporosis or cystic changes are seen in the femoral head on plain radiographs or CT scan. Still there is no evidence of subchondral fracture, fracture in the necrotic portion or flattening of the femoral head.	4	X-ray osteoarthritis		Osteoarthritis of the hip joint with joint space narrowing, acetabular changes and destruction are seen on plain radiographs.

出處 林松彥 骨質疏鬆與骨壞死

71

Which of the following statements is correct?

- (A) The main pathogenesis of osteoporosis is the imbalance of bone modeling.
- (B) The lumbar spine and bilateral hips are common sites for testing bone density.
- (C) A 65-year-old woman got a slip falling while walking and suffered a fracture of the first lumbar vertebra (L1 compression fracture). The DXA examination showed that T-score of the female neck is -1.2 and the average T-score of L2-L4 is -1.5. The woman was not eligible for the diagnosis of osteoporosis.
- (D) It is necessary to careful survey for coexisting problems or diseases with osteoporosis, when the T-score less than 2.0 was noted in young patients.

答案

(B) The lumbar spine and bilateral hips are common sites for testing bone density.

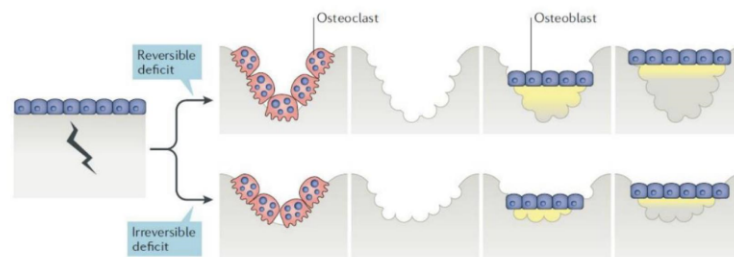
詳解

(A) 應改為bone remodeling

九、Pathophysiology 骨質疏鬆的病生理學

李怡臻 編撰

1. Unbalanced remodeling 骨重塑失衡



(1) Reversible deficit 正常情況：

蝕骨細胞(osteoclast)分解損壞的骨頭
→成骨細胞(osteoblast)將被分解出的坑洞修復補滿

(2) Irreversible deficit: unbalanced remodeling 骨重塑失衡：

蝕骨細胞(osteoclast)分解損壞的骨頭
→成骨細胞(osteoblast)無法填滿被分解出的坑洞
→未填滿的坑洞邊緣突起的部分會被吸收掉，骨頭回復平整
→惡性循環，每次骨頭代謝都無法補滿坑洞，造成骨量越來越少

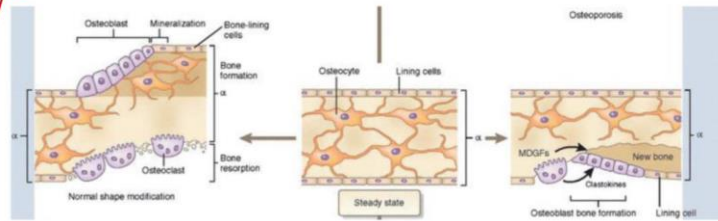
	(B)正確 3. 測量位置： (1) 腰椎或兩側髖關節，兩者都測更好 (2) 若上述兩處不能測時（如：開過刀或骨折而變形時） 可測「非慣用側」前臂橈骨 1/3 處(1/3- radius of non-dominant hand) (3) 補原理：上述部位骨密度相對較低，骨質疏鬆在該處表現較明顯，是骨質疏鬆性骨折好發位置。（非慣用手的骨密度往往比慣用手低） (M107) (C)走路時滑倒骨折屬於低創傷性骨折，因此即使T-score(-1.5)>-2.5，也可以診斷為骨質疏鬆。 2. 依據病史： (1) 低創傷性骨折(low traumatic fracture)或脊椎骨折病史，即使骨密度 T 值> -2.5 也得診斷為骨質疏鬆。 (2) Fragility fracture, particularly at the spine, hip, wrist, humerus, rib, and pelvis. 低創傷性骨折(low traumatic fracture)：從身高以下的高度掉下去，造成脊椎、髖關節、手腕、肋骨、肋骨或骨盆骨折。 (D)應改為Z-score<-2時需careful survey。 (2) Z-score A. 和同年齡層的標準差比較，不用來診斷。 B. A Z-score of -2 or lower is considered below the expected range for age. 當 Z-score < -2 的時候須注意。 C. Careful scrutiny for coexisting problems – Secondary osteoporosis 當 Z-score 較同年齡層低時，代表有其他疾病造成骨質疏鬆，須找出 Secondary osteoporosis 的成因 D. 補 (M107) Z-score 通常用在 40 歲以下。 E. 補 (M107) Z-score= $\frac{\text{BMD patient}-\text{BMD age matched reference}}{\text{SD age matched reference}} = \frac{\text{骨密度}-\text{正常同齡者 BMD}}{\text{正常同齡者的標準差}}$
出處	林松彥 骨質疏鬆與骨壞死

72	Which of the following statements is correct? (A) Bone modeling must have osteoclast and osteoblast coupling (B) RANK ligand inhibitor binds to RANK on osteoclasts to inhibit the maturation of osteoclasts (C) Continuous administration of Parathyroid hormone increase bone formation (D) Anti-sclerostin antibody promotes Wnt signaling pathway to increase bone synthesis and inhibits RANKL-dependent pathway to decrease bone breakdown
答案	(D) Anti-sclerostin antibody promotes Wnt signaling pathway to increase bone synthesis and inhibits RANKL-dependent pathway to decrease bone breakdown

詳解

(A) Remodeling 才會有先後順序之分

Modeling vs Remodeling

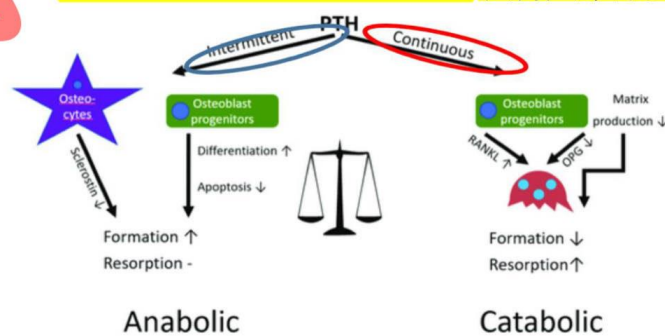


1. Modeling (左): osteoblast 及 osteoclast 可同時活動，因此骨頭不會有中間短暫流失的狀況 (小時候成長過程中)
2. Remodeling (右): osteoblast 及 osteoclast 有先後順序，中間會有短暫骨質流失的空窗期

(B) RANKL 應該為促進嗜骨細胞的成熟和作用

(C) 可參考下圖，間斷性給藥才能促進骨質

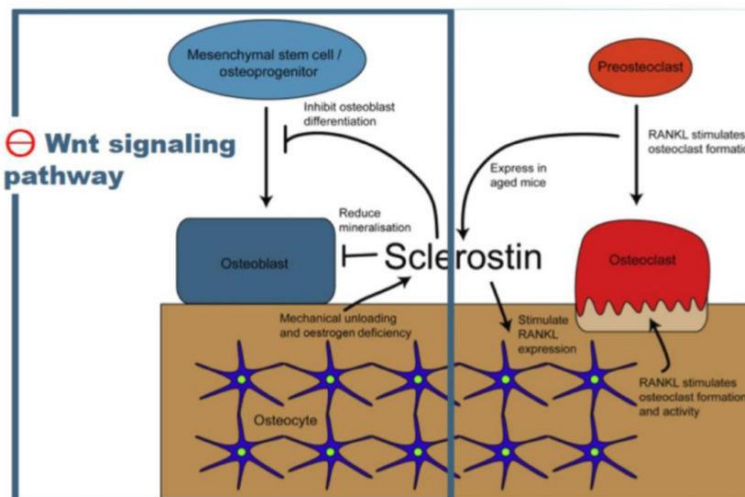
二十三、Effect of PTH on Bone Metabolism PTH 在骨質新陳代謝的作用



1. PTH 為可以促進骨質生成(Anabolic)的藥物。
2. PTH analog 的效果會依使用方式而不同
 - (1) Intermittent (間歇性的): 可以促進骨頭生長
 - (2) Continuous (每天打一針): 抑制骨頭生長

A. 因活化的 osteoblast 開始分泌大量 RANKL，使得嗜骨細胞活化

(D) 正確，左邊是 Wnt pathway，右邊是 RANKL 的 pathway



出處

林松彥 骨質疏鬆與骨壞死

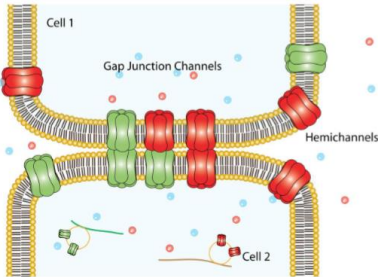
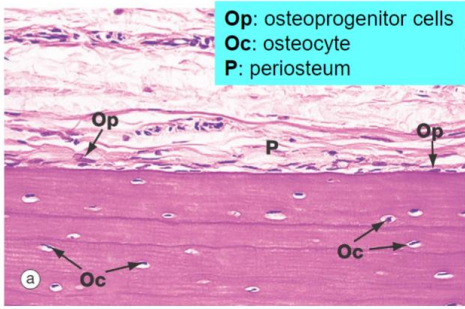
73	The key for management of neck and low back pain to find out pain generators. Which is not the potential pain generator of neck and lower back problems? (A) Bone (B) Ligament (C) Cartilage (D) Nerve root
答案	(C) Cartilage
詳解	Innervated Structures • Bone: vertebrae • Joints: zygapophyseal • Disk: only the external annulus and potentially diseased disk • Ligaments: anterior longitudinal ligament, posterior longitudinal ligament, interspinous • Muscles and fascia • Nerve root Noninnervated Structures • Ligamentum flavum • Disk: internal annulus, nucleus pulposus 疼痛來源無cartilage
出處	李佳玲 骨科復健學(1)

74	To control pain and inflammation under neck and lower back problems, which method is not suggested? (A) Bed rest till pain is gone (B) Oral medication (C) Exercise after acute stage (D) Manual therapy with massage
答案	(A) Bed rest till pain is gone

詳解	THERAPEUTIC INTERVENTIONS • Controlling pain and inflammation – <i>Pharmacologic management</i> • Oral medication • Interventional treatments – <i>Nonpharmacological management</i> • Physical • Behavioral • Regain normal function : – Education about the injury, the treatment objectives, and the prognosis Oral Medications • <i>Nonsteroidal Antiinflammatory Drugs</i> • <i>Muscle Relaxants</i> • <i>Antidepressants</i> • <i>Opioids</i> • <i>Antiseizure Medications</i> • <i>Tramadol</i> • <i>Systemic Steroids</i> • <i>Herbal Medicines</i> • <i>Topical Treatments</i> Nonpharmacological Managements • <i>Cervical and Lumbar Supports</i> • <i>Traction</i> • <i>Transcutaneous Electrical Nerve Stimulation and Other Modalities</i> • <i>Massage</i> • <i>Manual Mobilization or Manipulation</i> • <i>Reassurance and Patient Education (Back Schools)</i> • <i>Exercise</i> – <i>Therapeutic exercise</i> – <i>Aquatic Exercises</i> – <i>Traditional Exercise and Complementary Movement Therapies</i> 
出處	李佳玲 骨科復健學(1)

75	The pressure over disc changes in different positions. In which position the disk will suffer from the least pressure? (A) Side lying (B) Supine lying (C) Standing (D) Sitting
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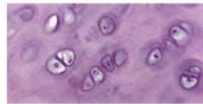
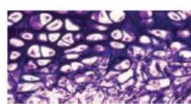
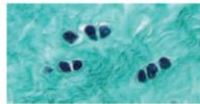
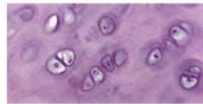
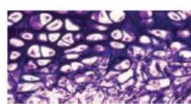
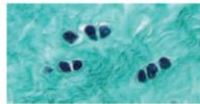
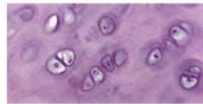
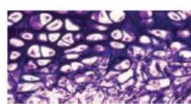
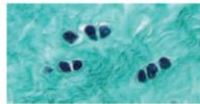
答案	(B) Supine lying
詳解	椎間盤受到的壓力 Sitting > Standing > Side lying > Supine lying , 特別注意壓力坐大於站。
出處	李佳玲 骨科復健學(1)

76	Which structure can make osteocytes exchange material with the extracellular matrix? (A) tight junction (B) adhering junction (C) hemichannel (D) gap junction
答案	(C) hemichannel
詳解	Osteocytes ❑ When osteoblasts completely surrounded by osteoid or bone matrix (3 days) ❑ Produce many extracellular matrix (ECM), reduce cell volume, decrease size and number of organelles, develop long <u>cell processes</u> ❑ Lacuna: each osteocyte occupied after bone mineralization ❑ Canaliculi: enclose the cell processes of osteocytes ❑ Hemichannels provide communication between cells and ECM • H&E stain→canaliculi and cell process can not be observed clearly   如左圖所示
出處	施耀翔 骨骼組織學(ptt pg. 42)

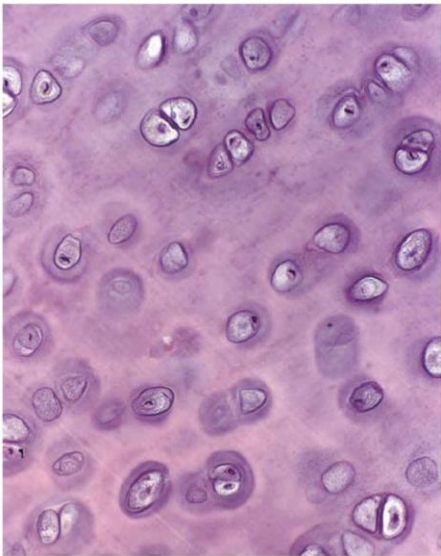
77	Which statement is correct about osteoclasts? (A) Can be found in Howship's lacunae (B) Single nucleus (C) Flatten shape of the nucleus (D) Developing from mesenchymal stem cells
答案	(A) Can be found in Howship's lacunae

詳解	Osteoclasts Morphology: large, multinucleated cells, acidophilia Location: where bone is being removed Resorption bay (Howship's lacuna) - Derived from the fusion of mononuclear hemopoietic progenitor cells (granulocyte/macrophage progenitor cells) (B) 錯，是多核 (C) 錯，不是flatten (D) 錯，是mononuclear hemopoietic progenitor cells
出處	施耀翔 骨骼組織學

78	The secondary ossification center of long bone is localized in_____. (A) diaphysis (B) epiphysis (C) metaphysis (D) endosteum
答案	(B) epiphysis
詳解	Endochondral ossification □ Begin at about 12th week ■ Epiphyseal cartilage 1. hyaline cartilage 2. perichondrium 分化成periosteum, 逐漸分裂形成osteoblasts, 在骨骼表面形成一團bone collar 3. matrix 鈣化, chondrocytes 死亡, 留下lacuna cavity 4. 血管進入空腔並帶入osteogenic cells 5. 原有的軟骨不斷被分解, 而硬骨化的部分擴大, 稱為ossification center 6,7. 出生前後, 新生血管會進入骨骼兩端的epiphysis, 其中一端先形成secondary ossification center 8. 另一端也形成secondary ossification center 9. 外圍留下一圈articular cartilage, 中央則留下生長板(growth plate) 10. 生長板會逐漸骨化癒合, 留下epiphyseal line
出處	施耀翔 骨骼組織學(ppt pg. 64)

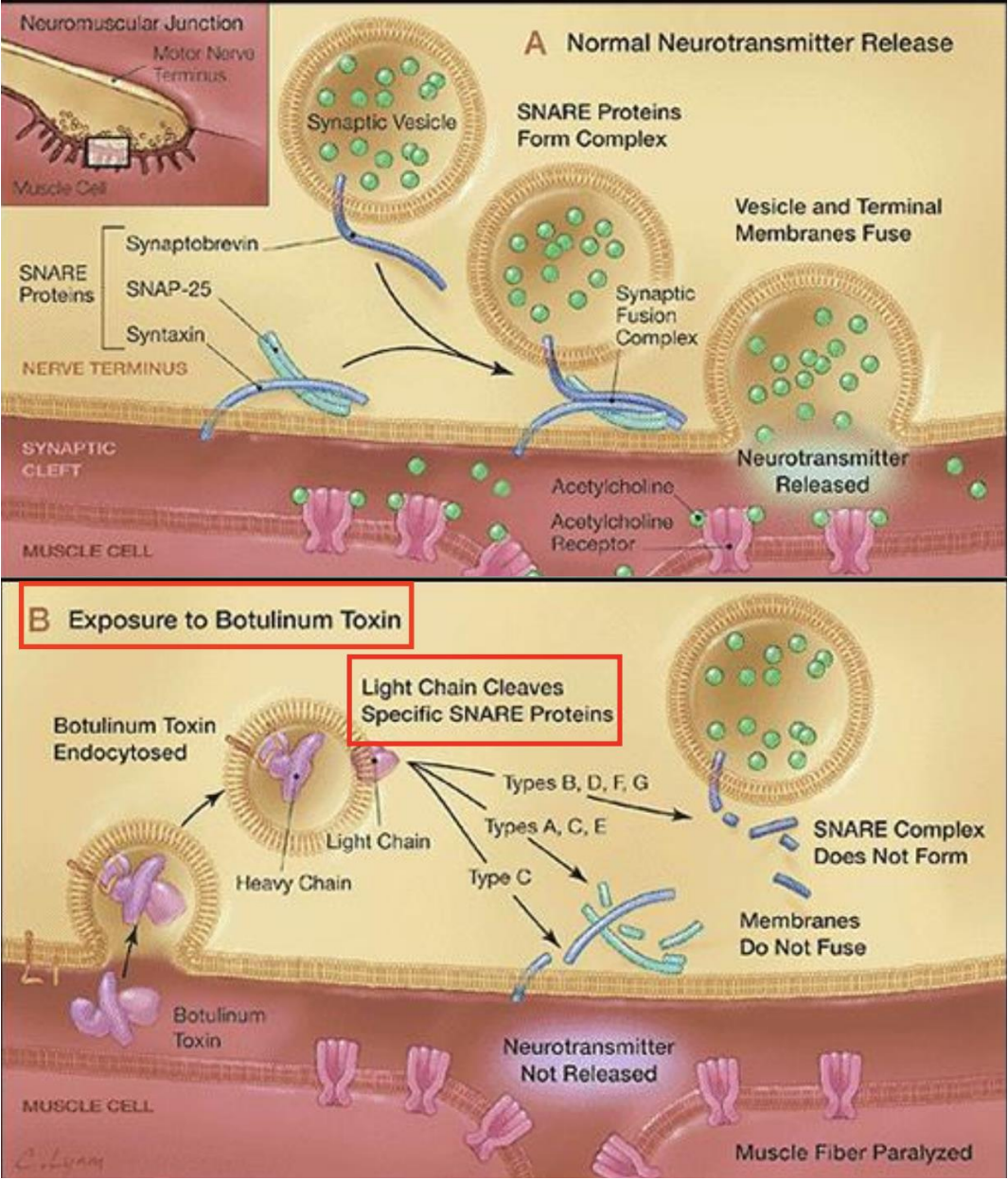
79	Which statement about cartilage is correct? (A) Elastic cartilage containing elastic fiber without any aggrecan protein. (B) The major type of collagen fiber is type II in hyaline cartilage. (C) The major type of collagen fiber is type II in fibrocartilage. (D) The major component of the extracellular matrix in all types of cartilage is collagen fiber.																																								
答案	(B) The major type of collagen fiber is type II in hyaline cartilage.																																								
詳解	(A) 彈性軟骨(elastic cartilage)，有彈性纖維(elastic fiber) 和蛋白聚糖(aggrecan) protein (B) 透明軟骨(hyaline cartilage)中的膠原纖維，主要為第二型 (C) 纖維軟骨(fibrocartilage)中的膠原纖維，主要為第一型 (D) 三種軟骨的胞外間質(extracellular matrix)的組成中，皆以水分所占比例為最高 TABLE 7.1 Summary of Cartilage Features <table><tr><th>Features</th><th>Hyaline Cartilage</th><th>Elastic Cartilage</th><th>Fibrocartilage</th></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td>Location</td><td>Fetal skeletal tissue, epiphyseal plates, articular surface of synovial joints, costal cartilages of rib cage, cartilages of nasal cavity, larynx (thyroid, cricoid, and arytenoids), rings of trachea, and plates in bronchi</td><td>Pinna of external ear, external acoustic meatus, auditory (Eustachian) tube, and cartilages of larynx (epiglottis, corniculate, and cuneiform cartilages)</td><td>Intervertebral discs, pubic symphysis, articular discs (sternoclavicular and temporomandibular joints), menisci (knee joint), triangular fibrocartilage complex (wrist joint), and insertion of tendons</td></tr><tr><td>Function</td><td>Resists compression Provides cushioning, smooth, and low-friction surface for joints Provides structural support in respiratory system (larynx, trachea, and bronchi) Forms foundation for development of fetal skeleton and further endochondral bone formation and bone growth</td><td>Provides flexible support for soft tissues</td><td>Resists deformation under stress</td></tr><tr><td>Presence of perichondrium</td><td>Yes (except articular cartilage and epiphyseal plates)</td><td>Yes</td><td>No</td></tr><tr><td>Undergoes calcification</td><td>Yes (i.e., during endochondral bone formation, during aging process)</td><td>No</td><td>Yes (i.e., calcification of fibrocartilaginous callus during bone repair)</td></tr><tr><td>Main cell types present</td><td>Chondroblasts and chondrocytes</td><td>Chondroblasts and chondrocytes</td><td>Chondrocytes and fibroblasts</td></tr><tr><td>Characteristic features of extracellular matrix</td><td>Type II collagen fibrils and aggrecan monomers (the most important proteoglycan)</td><td>Type II collagen fibrils, elastic fibers, and aggrecan monomers</td><td>Types I and II collagen fibers Proteoglycan monomers: aggrecan (secreted by chondrocytes) and versican (secreted by fibroblasts)</td></tr><tr><td>Growth</td><td colspan="3">Interstitally and appositionally, very limited in adults</td></tr><tr><td>Repair</td><td colspan="3">Very limited capability, commonly forms scar, resulting in fibrocartilage formation</td></tr></table> → • Hyaline cartilage is characterized by matrix containing type II collagen fibers, GAGs, proteoglycans, and multiadhesive glycoproteins. • Elastic cartilage is characterized by elastic fibers and elastic lamellae in addition to the matrix material of hyaline cartilage. • Fibrocartilage is characterized by abundant type I collagen fibers as well as the matrix material of hyaline cartilage.	Features	Hyaline Cartilage	Elastic Cartilage	Fibrocartilage					Location	Fetal skeletal tissue, epiphyseal plates, articular surface of synovial joints, costal cartilages of rib cage, cartilages of nasal cavity, larynx (thyroid, cricoid, and arytenoids), rings of trachea, and plates in bronchi	Pinna of external ear, external acoustic meatus, auditory (Eustachian) tube, and cartilages of larynx (epiglottis, corniculate, and cuneiform cartilages)	Intervertebral discs, pubic symphysis, articular discs (sternoclavicular and temporomandibular joints), menisci (knee joint), triangular fibrocartilage complex (wrist joint), and insertion of tendons	Function	Resists compression Provides cushioning, smooth, and low-friction surface for joints Provides structural support in respiratory system (larynx, trachea, and bronchi) Forms foundation for development of fetal skeleton and further endochondral bone formation and bone growth	Provides flexible support for soft tissues	Resists deformation under stress	Presence of perichondrium	Yes (except articular cartilage and epiphyseal plates)	Yes	No	Undergoes calcification	Yes (i.e., during endochondral bone formation, during aging process)	No	Yes (i.e., calcification of fibrocartilaginous callus during bone repair)	Main cell types present	Chondroblasts and chondrocytes	Chondroblasts and chondrocytes	Chondrocytes and fibroblasts	Characteristic features of extracellular matrix	Type II collagen fibrils and aggrecan monomers (the most important proteoglycan)	Type II collagen fibrils, elastic fibers, and aggrecan monomers	Types I and II collagen fibers Proteoglycan monomers: aggrecan (secreted by chondrocytes) and versican (secreted by fibroblasts)	Growth	Interstitally and appositionally, very limited in adults			Repair	Very limited capability, commonly forms scar, resulting in fibrocartilage formation		
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出處	施耀翔 骨骼組織學(ppt pg. 101, 103)																																								

80	Which of the following laryngeal cartilages does not have the structure of elastic cartilage? (A) Cricoid cartilage (B) Thyroid cartilage (C) Epiglottic cartilage (D) Arytenoid cartilage
答案	(A) Cricoid cartilage (B) Thyroid cartilage (D) Arytenoid cartilage
詳解	- Cricoid cartilage does not contain elastic cartilage. - It is made up of hyaline cartilage. - Furthermore, it's a complete and signet ring-shaped cartilage which is situated below the thyroid cartilage.
申覆	原因： 1. 根據老師講義第103頁表格（以及ROSS中文版第206頁表格），可在透明軟骨部分看到有Thyroid, Cricoid, Arytenoid三種軟骨，而可在彈性軟骨部分看到epiglottic cartilage，若依表格選擇應為ABD選項，且根據NIH資料整理顯示 (https://www.ncbi.nlm.nih.gov/books/NBK553185/)："The thyroid cartilage is the largest cartilage of the larynx and is composed of hyaline cartilage." 以及The cricoid is the only cartilage in the larynx that completely encircles the airway. It is composed of hyaline cartilage and resembles a signet ring in shape." 兩者寫法皆表示此兩軟骨為透明軟骨，另外此題為97-2牙醫國考題，當年度答案雖為A選項，但學生找不到資料佐證其想法，根據以上原文書及NIH只能確定C選項有含彈性軟骨，所以學生才認為答案為ABD，謝謝老師 2. 根據老師講義第103頁表格中顯示含透明軟骨之喉部構造為thyroid, cricoid, arytenoids，含彈性軟骨之喉部構造有epiglottis, corniculate, cuneiform cartilages)，故學生認為此題答案應為ABD，請老師斟酌給分，謝謝老師！ 申覆結果： A→ABD 老師回覆： 本題確實為97年度牙醫國考題更改，根據醫學系國考用書 Histology : A text and atlas, by Michael H. Ross, and Wojciech Pawlina. Lippincott / Williams & Wilkins. 答案應為ABD無誤。
出處	施耀翔 骨骼組織學

81	Which of the following is a characteristic of hyaline cartilage? (A) Can be found in the ear (B) Avascular (C) Concentric lamellae (D) Developed from mesenchymal stem cell
答案	(B) Avascular
詳解	(A) Elastic cartilage(彈性軟骨)比較會出現在耳朵附近的構造，例如external ear/ walls of the external acoustic meatus (外耳/外耳道壁)、auditory (Eustachian) tube(耳咽管)、epiglottis (會厭軟骨) of the larynx (B) 軟骨是一種無血管組織；胞外間質密實穩固，但同時也柔韌，造成其具有彈性。由軟骨細胞(chondrocytes)和大量分佈的胞外間質(extracellular matrix)組成 → 選項正確 OVERVIEW OF CARTILAGE • Cartilage is an avascular tissue that consists of chondrocytes and an extensive extracellular matrix. • Extracellular matrix in cartilage is <u>solid</u> and <u>firm</u> but also somewhat pliable (柔韌), which accounts for its resilience. • Glycosaminoglycans (GAGs) & type II collagen fibers.  Hyaline cartilage (C) Concentric lamellae 出現在cortical bone非hyaline cartilage (D) Hyaline cartilage主要由Chondrocyte組成，而Chondrocyte來自於neural crest cells(神經脊細胞) 分化過程如下：neural crest cells(神經脊細胞)→ ectomesenchyme(外胚層間質) → chondrogenic nodule→ chondroblasts → Chondrocyte (在軟骨節形成軟骨母細胞的過程中，會受到SOX9 的轉錄因子調節)

	• Chondrogenesis neural crest cells → ectomesenchyme → chondrogenic nodule → chondroblasts → Chondrocyte SOX9 ↑ • Cartilage has limited ability for repair. • When hyaline cartilage calcifies, it is replaced by bone.
出處	施耀翔 骨骼組織學(ppt pg. 101, 103)

82	Drugs that act on the central nervous system to induce skeletal muscle relaxation include the following drugs, except which one? (A) Central alpha-2 agonists (B) Depolarizing muscle relaxants (C) Tricyclic antidepressants (D) Antihistamines
答案	(B) Depolarizing muscle relaxants
詳解	Depolarizing agents 是 peripheral acting drug <pre> graph TD MR[Muscle relaxants] --> DA[Direct acting] MR --> CA[Centrally acting] MR --> PA[peripheral acting] DA --> D["• Dantrolene"] CA --> CA_List["CNS depressants • antihistamine • orphenadrine • sedatives • carisprodol • chlorzoxazone • metaxalone • methocarbamol • TCA-like • cyclobenzaprine • central alpha-2 agonists • tizanidine • GABA-agonists • Baclofen • benzodiazepines Botulinum toxin"] PA --> PA_List["Botulinum toxin Non-depolarizing agents Atracurium Alcuronium Cisatracurium Gallamine Pancuronium Rocuronium bromide Tubocurarine Vecuronium Depolarizing agents Suxamethonium"] </pre>
出處	程廣義 肌肉鬆弛劑之藥理學(ppt pg. 38)

83	Botox (<i>Clostridium botulinum</i>) acting which of the following to induce muscle relaxation? (A) Gamma-aminobutyric acid(GABA) type B receptor (B) Snare protein (C) Muscarinic receptors (D) Ligand-gated sodium channels
答案	(B) Snare protein
詳解	肉毒桿菌毒素會透過裂解snare protein 導致囊泡無法釋放Ach  A Normal Neurotransmitter Release Neuromuscular Junction Motor Nerve Terminus Muscle Cell Synaptic Vesicle SNARE Proteins Form Complex Vesicle and Terminal Membranes Fuse Synaptic Fusion Complex Neurotransmitter Released Acetylcholine Acetylcholine Receptor SYNAPTIC CLEFT MUSCLE CELL NERVE TERMINUS SNARE Proteins: Synaptobrevin, SNAP-25, Syntaxin B Exposure to Botulinum Toxin Botulinum Toxin Endocytosed Light Chain Cleaves Specific SNARE Proteins Types B, D, F, G Types A, C, E Type C SNARE Complex Does Not Form Membranes Do Not Fuse Neurotransmitter Not Released Muscle Fiber Paralyzed Botulinum Toxin Heavy Chain Light Chain
出處	程廣義 肌肉鬆弛劑之藥理學

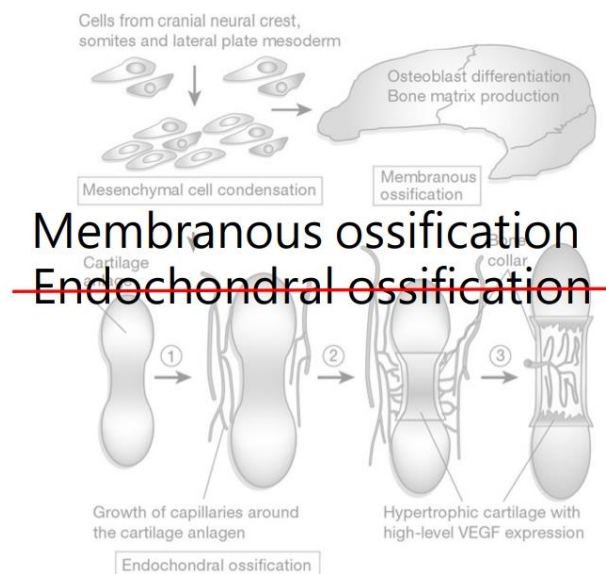
84	Characteristics of muscle pain, relative to cutaneous pain, include all of the following except (A) Poorly localized (B) Burning and cutting (C) Easily pain referral (D) Difficult to tolerate												
答案	(B) Burning and cutting												
詳解	TABLE Subjective differences between muscle pain and cutaneous pain <table> <tr> <th>Muscle pain</th><th>Cutaneous pain</th></tr> <tr> <td>Electrical nerve stimulation induces only one pain</td><td>Electrical nerve stimulation induces a first pain and a second pain</td></tr> <tr> <td>Poorly localizable</td><td>Well-localized</td></tr> <tr> <td>Tearing, cramping, pressing quality</td><td>Stabbing, burning, cutting quality</td></tr> <tr> <td>Marked tendency toward referral of pain</td><td>No tendency toward referral of pain</td></tr> <tr> <td>Affective aspect: difficult to tolerate</td><td>Affective aspect: easier to tolerate</td></tr> </table> 由表格可以看到muscle pain是tearing, cramping, pressing quality ; cutaneous pain才是 Burning and cutting。	Muscle pain	Cutaneous pain	Electrical nerve stimulation induces only one pain	Electrical nerve stimulation induces a first pain and a second pain	Poorly localizable	Well-localized	Tearing, cramping, pressing quality	Stabbing, burning, cutting quality	Marked tendency toward referral of pain	No tendency toward referral of pain	Affective aspect: difficult to tolerate	Affective aspect: easier to tolerate
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Poorly localizable	Well-localized												
Tearing, cramping, pressing quality	Stabbing, burning, cutting quality												
Marked tendency toward referral of pain	No tendency toward referral of pain												
Affective aspect: difficult to tolerate	Affective aspect: easier to tolerate												
出處	程廣義 肌肉鬆弛劑之藥理學(ppt pg. 7)												

85	Which of the following descriptions about achondroplasia is correct? (A) The issue arises from the fourth pair of chromosomes, and it follows an autosomal dominant inheritance pattern. (B) Typically, patients have short limbs while spinal length is normal. (C) Patients lack normal membranous ossification during bone development. (D) It is associated with activating mutations in the FGFR3 gene.
答案	(C) Patients lack normal membranous ossification during bone development.
詳解	(A)和(B)



體染色體顯性遺傳
第4對染色體(4p16.3)
脊椎長度正常、四肢短

(C)



(D)

Activating mutation
Fibroblast Growth Factor Receptor 3

出處

陳建亨 肌肉骨骼病理學(1)(ppt pg. 26, 27, 28)

86

Which of the following descriptions about osteopetrosis is incorrect?

- (A) Patients have bones as hard as marble, which makes them less prone to fractures.
- (B) The main course is the inability to generate osteoclasts or the loss of bone resorption ability in generated osteoclasts.
- (C) In addition to skeletal abnormalities, patients often exhibit conditions such as cranial nerve paralysis, recurrent bacterial infections, or hepatosplenomegaly.
- (D) Patients can undergo hematopoietic stem cell transplantation for treatment.

答案	(A) Patients have bones as hard as marble, which makes them less prone to fractures.
詳解	(A) 骨頭脆弱容易骨折  
出處	陳建亨 肌肉骨骼病理學(1)(ppt pg. 36, 37)

87	Which of the following descriptions about osteogenesis imperfecta is incorrect? (A) Patients cannot produce normal type I collagen. (B) Mutations in the 17q21.3-q22 region, COL1A1 gene, and 7q21.3-q22 region, COL1A2 genes are associated with the condition. (C) It follows an autosomal dominant inheritance pattern and can be categorized into four types. (D) Due to the lack of normal type I collagen, patients' irises may exhibit Lisch nodules.
答案	(D) Due to the lack of normal type I collagen, patients' irises may exhibit Lisch nodules.
詳解	Type I collagen (88%) Blue sclerae Hearing loss Small misshapen teeth 玻璃娃娃眼睛會出現的是藍鞏膜(blue sclerae) 虹膜色素缺陷瘤(Lisch nodules)常見於I型神經纖維瘤病-課外知識 from wiki
出處	陳建亨 肌肉骨骼病理學(1)(ppt pg. 41-46)

88

Which of the following descriptions about fractures is most appropriate?

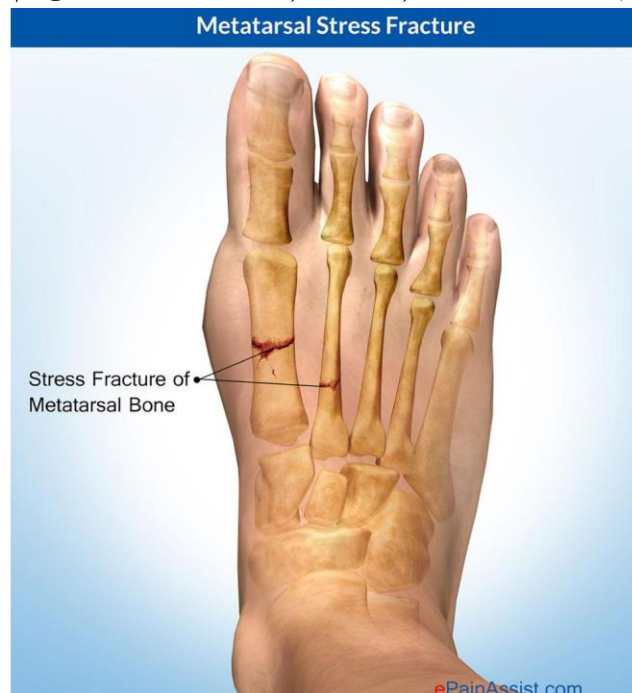
- (A) Stress fractures often occur in weight-bearing bones of the feet.
 (B) Fractures resulting from bone weakening due to disease are termed pathological fractures.
 (C) After approximately six weeks, the fracture site will show a complete meshwork of woven bone.
 (D) All of the above descriptions (a, b, c) are correct.

答案

(D) All of the above descriptions (a, b, c) are correct.

詳解

(A) Stress fracture: 壓力性骨折，骨骼因長時間受力而過於疲勞、裂開。常見於 metatarsals, tibia, navicular。容易發生於軍人、運動員。



(B) 骨頭被疾病破壞(如:代謝性疾病造成的骨質疏鬆)，使得骨頭強度較正常情況脆弱，只要稍加施力時便會骨折，此情形便稱為 pathological fractures

Pathological fractures

Bone weakened by disease

(C)

Table 9.1. Timetable for fracture healing (after Collins 1966)	
12 h	Blood clot and exudate between fragments
24 h	Acute inflammation with migration of polymorphonuclears and macrophages
48 h	Formation of granulation tissue
5 days	Earliest new bone formation
7 days	Empty osteocyte lacunae in remaining dead fragments
3 weeks	Fibrous union; some primary callus
6 weeks	Periosteal shell of external callus; complete meshwork of woven bone
After 6 weeks	Progressive formation of secondary callus and subsequent remodelling

出處	陳建亨 肌肉骨骼病理學(1)(ppt pg. 49, 50, 52)
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89	Which of the following statements about osteonecrosis (avascular necrosis) is incorrect? (A) It is mostly caused by infections. (B) It is associated with trauma, alcohol abuse, or steroid use. (C) In patients with prolonged alcohol abuse, osteonecrosis often occurs in the femoral head. (D) Osteonecrosis typically first affects the medullary bone, sparing the overlying cortical bone and cartilage cap temporarily due to differences in blood supply.
答案	(A) It is mostly caused by infections.
詳解	造成骨壞死男性常見原因為喝酒, 女性常見原因為自體免疫疾病。  Osteonecrosis (Avascular necrosis; 骨壞死) 非感染性原因 外傷、酗酒、類固醇... 等

	三、Epidemiology 流行病學 10 percent of all total hip replacements (THR) 髖關節置換術患者的的 10% - Male > Female 男性多於女性 (1) 要記！在骨骼關節疾病中，只有少數的疾病是男性大於女性 (2) Alcohol-associated ON: male 男性常見原因：喝酒 A. 造成脂肪堆積於細小血管上，femoral head 最常見 (3) Systemic lupus erythematosus (SLE)- associated ON: females 女性常見原因：自體免疫疾病(SLE) A. 可能因服用類固醇而導致病情加重 - Usually less than 50 years 患者通常小於 50 歲 - Bilateral hips involved 80% of the time 80%的人雙側髖骨皆受影響 3% of patients with osteonecrosis have multifocal involvement 3%的人的疾病是多發性的，其他處的骨頭也有壞死
出處	陳建亨 肌肉骨骼病理學(1)(ppt pg. 56)

90	Which of the following descriptions about osteomyelitis is incorrect? (A) It is caused by bacterial infections either through direct exposure or bloodstream transmission. (B) In children, it often occurs near the knee joint, while in adults, it's more common in the feet. (C) Osteomyelitis near the metaphysis of long bones in 5-10-year-old children is often caused by common E. coli bacteria. (D) Osteomyelitis can occur in the maxillary and mandible bones, often related to poor oral hygiene and dental caries.
答案	(C) Osteomyelitis near the metaphysis of long bones in 5-10-year-old children is often caused by common E. coli bacteria.
詳解	5-10歲經血液至長骨metaphysis Staphylococcus aureus (金黃色葡萄球菌)
出處	陳建亨 肌肉骨骼病理學(1)(ppt pg. 63)